A Virtual Laboratory for Vertebrate Brain Research: Active Inference, Spiking Neural Networks, and Controlled Behavioral Experiments in a Biologically Grounded Zebrafish Brain Model

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Abstract

We present a spiking neural network (SNN) model of the larval zebrafish brain that implements a complete perception–action loop under the active inference framework. The architecture comprises 48 brain modules totalling 7,316+ Izhikevich spiking neurons with conductance-based synapses (AMPA, NMDA, GABA_A), organized into 39 spiking and 9 rate/analytic subsystems. Seven cell types (RS, IB, FS, CH, LTS, TC, MSN) populate an excitatory–inhibitory balanced network (75% E / 25% I) spanning retina, optic tectum (8 hemispheric layers, 3 200 neurons with full optic-chiasm decussation), thalamo-pallial loop (2 780 neurons with predictive coding and top-down tectal attention), basal ganglia, reticulospinal system, and a 32-neuron LIF spinal central pattern generator. A spiking saccade module implements active-inference gaze control, directing eye position toward food, predators, or high-prediction-error hemifields. Goal selection minimizes Expected Free Energy (EFE) across four policies (forage, flee, explore, social), computed from spiking classifier output, spiking amygdala fear signal, cerebellar prediction error, olfactory gradients, circadian activity drive, predictive surprise, and an anticipatory energy-anxiety signal that drives foraging preference in the middle energy range ($\sim$40–65%), resolved by a spiking winner-take-all attractor whose projection weights are trained by reward-modulated three-factor Hebbian learning. The model includes a spiking RL critic with TD learning and eligibility traces (with DA-gated STDP consolidation), a VAE world model with batched ELBO training and associative memory planning, spiking olfaction with alarm substance, spiking lateral line, color vision, circadian clock, sleep–wake regulation, and four additional sensory modalities (nociception, auditory, gustatory, tactile), hypothalamus, pineal gland, inferior olive, dorsolateral pallium (hippocampal homolog), vagus nerve, pituitary, area postrema, nucleus of the solitary tract (NTS), and lateral line efferent system that complete the zebrafish sensorium. Four-axis neuromodulation (DA, NA, 5-HT, ACh) modulates plasticity, attention, and behavioral state. Online classifier fine-tuning reinforces both food (ACh-gated) and enemy (NA-gated) categories from behavioral confirmation signals. The reticulospinal voluntary pathway, driven by pallium-D left/right asymmetry gated through the basal ganglia, contributes to non-escape motor output alongside the primary retinal turn signal. Ablation reveals that olfaction is the only module whose removal reduces survival (-7%), while habenula and olfaction each reduce food intake by -35%, followed by RL critic (-28%) and amygdala (-25%). The system achieves 84/100 on a five-scenario decision rationality benchmark: safe food selection (A=78/100), predator-charge detection perfect (B=100/100), starvation dilemma (C=100/100), easy foraging (D=76/100), and exploration (E=68/100). Across 10 seeds, all reach the full 500-step episode (mean 500 ± 0 steps), consuming 11.1 ± 2.8 food items (0/10 caught). A multi-agent configuration with 5 fish—each with a full v2 brain (35,000 total spiking neurons) and individual personality profiles (bold, shy, explorer, social)—demonstrates emergent competitive foraging and collective predator avoidance against an intelligent 5-state predator AI. A retrained spiking classifier achieves 96.2% accuracy, a three-stage curriculum passes all stages, and prey capture kinematics (J-turn $\rightarrow$ approach swim $\rightarrow$ capture strike) improve foraging precision through binocular depth estimation. All learning uses biologically plausible local rules—DA-gated three-factor STDP with eligibility traces

($\tau_e = 500\,\mathrm{ms}$, consolidation only when DA > 0.55 or during phasic bursts), PE-driven anti-Hebbian feedback, TD-based value learning, and reward-modulated Hebbian updates to the goal-selection attractor and classifier output layer—without backpropagation through time. Each model neuron is positioned in 3D using coordinates from the MPIN whole-brain calcium imaging atlas (47,010 cells, 72 bilateral regions, 61 with recorded cells in subject 12) with connectivity derived from a reconstructed connectome (10.2 M synapses), making this the first anatomically constrained, fully spiking whole-brain model of a vertebrate. An interactive web platform supports real-time visualization via vtk.js, live brain surgery (ablation of 10 regions during execution), four-stage curriculum training, episode replay, and multi-agent schooling. Beyond its role as a neural model, the virtual zebrafish functions as a *virtual laboratory*: a controlled experimental environment in which researchers can test mechanistic hypotheses by manipulating specific biophysical parameters, neuromodulatory baselines, personality profiles, or social group composition, then observing emergent behavioral and neural consequences. Five scenario-based experimental protocols are provided: tabula rasa world learning, spatial amnesia recovery, fear conditioning acquisition, social inference calibration, and goal-bias adaptation under shifted reward statistics. The system further enables *disorder modeling*—producing hypodopaminergic, ASD-like (high serotonin / reduced social drive), and schizophrenia-like (NMDA hypofunction / DA dysregulation) behavioral phenotypes by adjusting neuromodulatory parameters and E/I balance, bridging computational psychiatry, behavioral science, and cognitive neuroscience within a single unified framework.

Keywords: active inference, predictive coding, spiking neural network, Izhikevich neuron, zebrafish, free energy principle, neuromodulation, reinforcement learning, virtual organism, excitatory-inhibitory balance

1 Introduction

Understanding how vertebrate brains generate adaptive behavior from neural circuits remains a central challenge in computational neuroscience. While substantial progress has been made in characterizing individual brain regions and their functions, a critical gap persists between abstract computational models of cognition and biologically realistic neural simulations. Abstract models of active inference (Friston, 2010; Parr et al., 2022) and predictive coding (Rao and Ballard, 1999; Bogacz, 2017) provide elegant normative accounts of brain function but are typically formulated at a level of abstraction far removed from specific neural circuits. Conversely, detailed biophysical models of spiking neurons capture cellular-level dynamics but rarely scale to the complete sensorimotor loops that generate behavior.

The larval zebrafish (*Danio rerio*, 5–7 days post-fertilization) offers an ideal model system for bridging this gap. Its brain contains approximately 100,000 neurons—small enough for whole-brain calcium imaging at cellular resolution (Ahrens et al., 2013)—yet generates a rich behavioral repertoire including visually guided prey capture (Bianco and Engert, 2015), predator avoidance with Mauthner cell escape reflexes (Temizer et al., 2015; Dunn et al., 2016), social behavior and shoaling (Dreosti et al., 2015), and optomotor responses (Portugues et al., 2014). The transparency of the larval brain enables simultaneous recording of neural activity across all major brain regions during behavior, providing uniquely comprehensive datasets for constraining computational models. Furthermore, the zebrafish brain contains homologs of all major vertebrate structures—optic tectum (superior colliculus), pallium (cortex), habenula, basal ganglia, hypothalamus, and cerebellum—making it a tractable proxy for understanding general principles of vertebrate brain organization.

The active inference framework (Friston, 2010; Friston et al., 2017; Parr et al., 2022) proposes that all brain function can be understood as minimizing variational free energy—an upper bound on sensory surprise. Under this framework, perception corresponds to updating internal beliefs to minimize prediction errors (perceptual inference), action corresponds to changing sensory inputs to match predictions (active inference), and learning corresponds to optimizing the generative model parameters (perceptual learning). Goal-directed behavior emerges from minimizing *expected* free energy (EFE), which naturally balances pragmatic value (achieving preferred outcomes) with epistemic value (reducing uncertainty). This framework provides a principled account of how organisms simultaneously perceive, act, learn, and plan within a single information-theoretic objective.

In this paper, we present a computational model—the Virtual Zebrafish—that implements the active inference framework in a spiking neural network architecture inspired by the larval zebrafish brain. The model is constructed through a 43-step developmental curriculum, where each step introduces a new computational module corresponding to a biological brain structure or function. This developmental approach mirrors biological maturation, where neural circuits come online progressively, and enables systematic testing at each stage of complexity.

The key contributions of this work are:

1. A complete spiking neural network architecture with two-compartment predictive coding neurons, goal-driven attention, and homeostatic gain control that implements hierarchical free energy minimization.
2. Integration of active inference goal selection, reinforcement learning, and Hebbian plasticity within a single sensorimotor loop, demonstrating their complementary roles in generating adaptive behavior.
3. A 43-step developmental curriculum that systematically builds a virtual organism from retinal sampling to episodic spatial memory, with quantitative evaluation at each stage.
4. Demonstration that the free energy principle provides a unified computational language for exteroceptive perception, interoceptive regulation, neuromodulation, and motor control.
5. A *virtual laboratory* framework with five experimental scenario protocols, standardized behavioral assays, and parameter-manipulation APIs that allow researchers across neuroscience, cognitive science, behavioral science, and computational psychiatry to test

mechanistic hypotheses without animal experiments—including disorder phenotype generation (hypodopaminergic, ASD-like, schizophrenia-like) and group-level social phenomena (schooling, collective alarm propagation, competitive foraging).

6. **vzlab**: a multi-species virtual laboratory extending the active inference framework to *C. elegans* (302 neurons), *Drosophila melanogaster* (2,145-neuron mushroom body), and *Xenopus laevis* tadpole (150-neuron spinal CPG). All four species achieve Grade A^+ on a four-tier validation protocol (behavioural battery, atlas correspondence, lesion replication, and robustness under simulated neuron dropout). The zebrafish retains full behavioral coherence at 50% sensory neuron dropout (robustness index = 1.0), demonstrating that the cells-to-social-behavior hierarchy is inherently fault-tolerant.

The model achieves 84/100 on a decision rationality benchmark (round 177 checkpoint), demonstrates online learning that triples foraging efficiency, and maintains adaptive behavior under concurrent predator threat and energy depletion—all using exclusively local, biologically plausible learning rules. The multi-species platform validates that the same active inference architecture generalises across four phylogenetically distant organisms spanning 302 to 100,000 neurons.

Figure 1 provides a schematic overview of the complete system: the virtual environment with annotated sensory modalities (Panel A), the five-layer biologically grounded brain architecture (Panel B), the closed active inference perception–action loop (Panel C), and key benchmark results (Panel D).

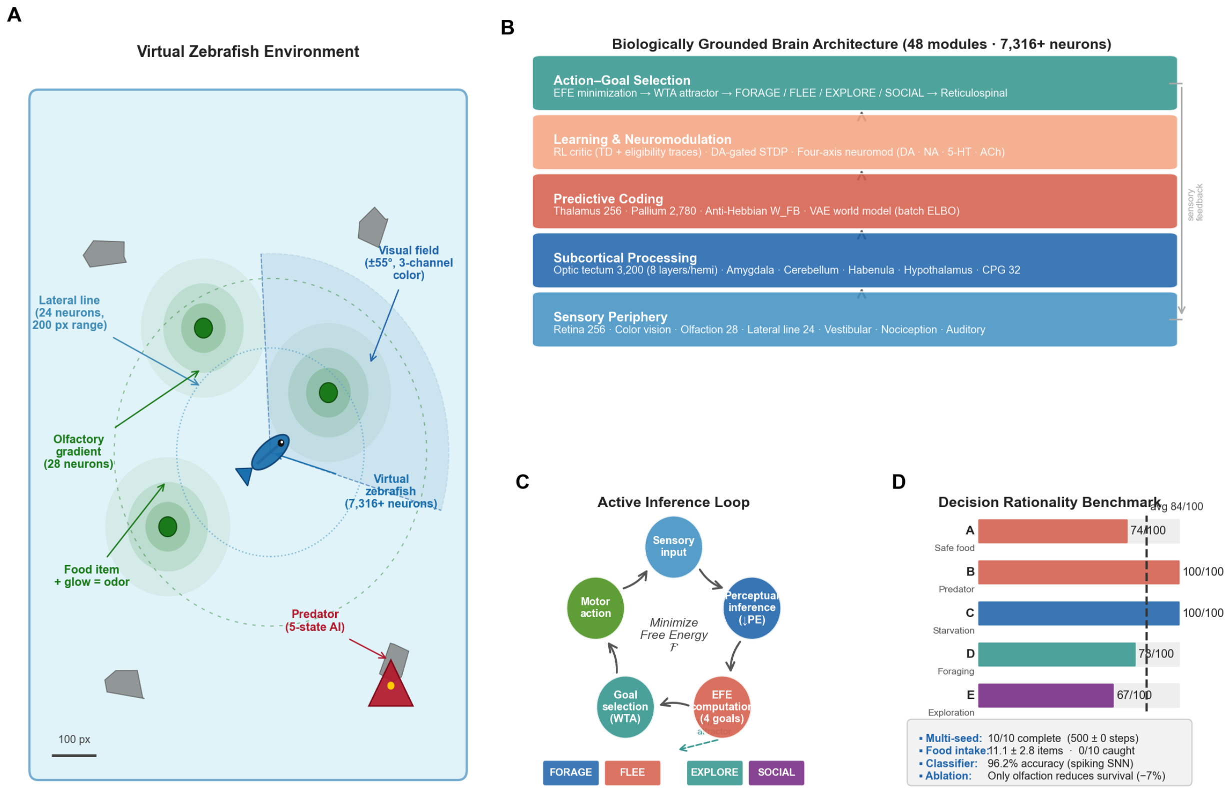

Figure 1: Overview of the Virtual Zebrafish system. **(A)** Behavioral arena: the fish (blue ellipse) navigates an 800×600 px environment containing food items (green), rocks (grey), and an intelligent 5-state predator (red). Annotated sensor modalities include the binocular visual field (±55ř, 3-channel color), the lateral-line mechanosensory ring (24 Izhikevich neurons, 200 px range), and the olfactory gradient field (28 neurons, Fick diffusion model). **(B)** Brain architecture: five functional layers—sensory periphery (retina 256 + olfaction + lateral line + color vision), subcortical processing (optic tectum 3,200 neurons, amygdala, cerebellum, habenula), predictive coding (thalamo-pallial loop 3,036 neurons with anti-Hebbian W_{FB}), learning and neuromodulation (DA-gated STDP, RL critic, four-axis neuromodulation), and action–goal selection (EFE minimisation, WTA attractor, reticulospinal pathway). **(C)** Active inference loop: perception minimises prediction error (PE), Expected Free Energy (EFE) is computed across four goal policies (FORAGE, FLEE, EXPLORE, SOCIAL), a winner-take-all attractor selects the active goal, and the resulting motor command closes the sensorimotor loop. **(D)** Decision rationality benchmark across five structured scenarios (A–E), with average784/100 (dashed line) and summary statistics from 10-seed evaluation.

2 Related Work

Zebrafish Neuroscience Whole-brain calcium imaging in larval zebrafish (Ahrens et al., 2013) revealed that visuomotor behavior engages distributed neural networks spanning the optic tectum, hindbrain, and cerebellum. Portugues et al. (2014) mapped stereotyped activity patterns during optomotor responses, identifying distinct functional modules for sensory processing, motor planning, and action execution. Prey capture in larval zebrafish involves a characteristic sequence of J-turns, approach swims, and capture strikes, with tectal neurons encoding prey position in a retinotopic map (Bianco and Engert, 2015). Predator avoidance relies on looming-sensitive neurons in the tectum that trigger Mauthner cell-mediated C-start escape reflexes when the angular expansion rate of an approaching object exceeds a threshold (Temizer et al., 2015; Dunn et al., 2016). Social behavior emerges early in development, with larval zebrafish showing attraction to conspecifics and modulation of activity by social context (Dreosti et al., 2015). The habenula plays a critical role in behavioral flexibility, encoding aversive expectation values that enable adaptive avoidance strategies (Amo et al., 2014; Jesuthasan and Bhatt, 2012).

Active Inference and Predictive Coding The free energy principle (Friston, 2010) proposes that biological systems minimize variational free energy—the divergence between their internal generative model and sensory observations. Predictive coding (Rao and Ballard, 1999) implements this minimization through a hierarchy of prediction errors, where each cortical layer predicts the activity of the layer below and updates its representation based on the mismatch. Bogacz (2017) provided a tutorial unification of predictive coding with Bayesian inference, showing that prediction error minimization approximates exact Bayesian updating under Gaussian assumptions. Clark (2013) extended this framework to an encompassing theory of brain function where prediction pervades all levels of neural processing. Active inference (Friston et al., 2017; Parr et al., 2022) generalizes predictive coding to action: organisms act to fulfill their generative model's predictions, selecting policies that minimize expected free energy. The EFE objective naturally decomposes into pragmatic (reward-seeking) and epistemic (uncertainty-reducing) components, providing a normative account of exploration-exploitation trade-offs. Attention has been formalized as precision optimization—the brain's allocation of gain to sensory channels in proportion to their expected reliability and task-relevance (Feldman and Friston, 2010). Interoceptive inference extends the framework to homeostatic regulation: organisms minimize prediction errors across both exteroceptive and interoceptive Markov blankets (Seth and Friston, 2016; Barrett and Simmons, 2015; Pezzulo et al., 2015).

Spiking Neural Networks and Virtual Organisms Biologically detailed spiking neural networks have been used to model specific neural circuits, with Spaun (Eliasmith et al., 2012) being a notable large-scale example that demonstrates cognitive tasks using 2.5 million leaky integrate-and-fire neurons. However, Spaun focuses on cortical function in primates rather than whole-organism behavior. The OpenWorm project (OpenWorm, 2014) aims to simulate *C. elegans* at cellular resolution but has not yet achieved complete behavioral repertoires. Merel et al. (2019) developed virtual rodent models using deep reinforcement learning to control articulated bodies, demonstrating realistic locomotion but without biological neural architectures. Animat approaches (Wilson, 1991) pioneered the idea of simulated organisms but typically used simplified neural networks without biological grounding.

Reinforcement Learning and Dopamine The correspondence between temporal-difference (TD) learning (Sutton and Barto, 2018) and dopamine neuron firing patterns (Schultz et al., 1997) established a foundational link between computational reinforcement learning and neurobiology. Actor-critic architectures map naturally onto basal ganglia circuits, with the ventral striatum encoding state values and the dorsal striatum implementing action selection (O'Doherty et al., 2004). Dabney et al. (2020) demonstrated that distributional reinforcement learning provides a better account of dopamine neuron diversity than scalar TD error. The integration of model-free RL (habitual behavior) with model-based planning (goal-directed behavior) has been extensively studied (Daw et al., 2011), with evidence that these systems compete and cooperate depending on uncertainty and computational cost.

Gap in the Literature Despite significant advances in each of these areas, no existing model combines a spiking neural network architecture with full active inference (perception, action, learning, and planning), a complete sensorimotor loop (from retina to motor output), emotional and interoceptive regulation, and multiple learning systems (genomic, Hebbian, reinforcement) within a single virtual organism operating in a complex ecological environment. The present work addresses this gap by building a comprehensive virtual zebrafish through a systematic developmental curriculum.

3 Methods

The virtual zebrafish brain comprises 48 functional modules, organized here by subsystem. Table 11 provides a complete inventory.

3.1 SNN Architecture

Izhikevich Spiking Neurons All spiking layers use the Izhikevich neuron model (Izhikevich, 2003) with midpoint integration at $\Delta t = 1\,\mathrm{ms}$, supporting seven cell types:

$$\frac{dv}{dt} = 0.04\,v^2 + 5v + 140 - u + I, \quad \frac{du}{dt} = a(bv - u), \tag{1}$$

with spike reset $v \to c,\ u \to u + d$ when $v \geq 30\,\mathrm{mV}$. Parameters (a, b, c, d) define seven cell types: regular-spiking (RS), intrinsically bursting (IB), fast-spiking (FS), chattering (CH), low-threshold spiking (LTS), thalamocortical (TC), and medium-spiny (MSN).

Excitatory–Inhibitory Layers Each cortical-like region uses an EI layer with 75% excitatory and 25% inhibitory neurons, connected by four conductance-based synapse classes (AMPA, NMDA, GABA_A):

$$I_{\text{syn}} = g_{\text{AMPA}}(E_E - v) + g_{\text{NMDA}} \frac{(E_E - v)}{1 + 0.28\, e^{-0.062v}} + g_{\text{GABA}}(E_I - v). \tag{2}$$

Each behavioral step runs 50 substeps ($\times 1$ ms), yielding 50 ms of neural dynamics per decision cycle.

Two-Compartment Predictive Coding Every module in the brain contains a *TwoCompColumn*: a micro-column of Izhikevich two-compartment neurons where prediction error is computed within each neuron via apical–somatic mismatch (Larkum, 2013; Lee, Lee & Park, 2026):

$$\boldsymbol{\varepsilon}^{(l)} = \mathbf{v}_a^{(l)} - \widetilde{\mathbf{v}}_s^{(l)}, \tag{3}$$

where the tilde denotes normalisation of the somatic potential to $[0, 1]$. Following Lee et al. (2026), the additive somatic bias serves as the precision modulation:

$$b_i = b_0 + \sigma(\gamma_i) \cdot g + \alpha_{\text{att}}\, m_{\text{att}}, \tag{4}$$

where γ_i is a learnable precision parameter updated online via $d\gamma/dt = \alpha_\pi(|\varepsilon| - \beta)$ (Bogacz, 2017). This yields brain-level variational free energy:

$$F_{\text{brain}} = \sum_{m=1}^{10} \sum_i \tfrac{1}{2}\, \sigma\big(\gamma_i^{(m)}\big) \left(\varepsilon_i^{(m)}\right)^2, \tag{5}$$

aggregating 292 two-compartment neurons across all sensory, thalamic, pallial, and motor modules (Table 1). Feedback weights W_{FB} learn via anti-Hebbian STDP: $\Delta W_{\text{FB}} = -\eta\, \mathbf{h}_{\text{upper}}^T\, \boldsymbol{\varepsilon}$.

Table 1: Two-compartment predictive coding columns across all modules. Each column contains Izhikevich neurons with apical (prediction) and somatic (evidence) compartments. PE $= V_a - \tilde{V}_s$ within each neuron; precision $\pi = \sigma(\gamma)$ learned online.

Module	Ch	Sensory (soma)	N
Retina	4	L/R ON & OFF frame rates	16
Olfaction	2	food & alarm concentration	8
Lateral line	3	flow anterior/posterior, canal threat	12
Proprioception	4	speed, heading, wall, collision	16
Vestibular	2	turn rate, speed	8
Color vision	4	UV, B, G, R spectrum	16
Nociception	3	mechanical, thermal, chemical	12
Auditory	3	low, mid, high frequency	12
Gustatory	3	amino acid, bitter, umami	12
Tactile	3	head, trunk, tail	12
Hypothalamus	5	LH, VMH, PVN, POA, SCN	20
Pineal	2	parapinopsin, exo-rhodopsin	8
Inferior olive	2	climbing fiber, gap-junction	8
Dl pallium (hippocampus)	4	DG, CA3, CA1, SWR	16
Vagus nerve	2	visceral afferent, cardiac PE	8
Pituitary	3	anterior, intermediate, posterior PE	12
Area postrema	2	chemosensory, metabolic PE	8
NTS	2	gustatory, visceral PE	8
Lateral line efferent	2	flow prediction, suppression PE	8
Thalamus ×2	2	TC relay, TRN gate	16
Pallium	2	superficial, deep rates	8
Active motor	8	8 proprioceptive channels	48
Total	**69**		**292**

Processing Pipeline The hierarchical path through the v2 architecture is:

$$\underbrace{\text{Retina}}_{\text{4 RGC}\times 2} \rightarrow \underbrace{\text{Tectum}}_{\text{8 hemi. layers (3200)}} \rightarrow \underbrace{\text{Thalamus}_{L/R}}_{\text{TC+TRN (380)}} \rightarrow \underbrace{\text{Pallium}}_{\text{S+D (2400)}} \rightarrow \underbrace{\text{BG}}_{\text{D1/D2/GPi (760)}} \rightarrow \underbrace{\text{RS}}_{\text{21/side}} \rightarrow \underbrace{\text{CPG}}_{\text{32 LIF}} \tag{6}$$

Goal-Driven Attention Modulator Eight attention neurons (two per goal) with slow neuromodulatory dynamics ($\tau_{\text{att}} = 0.93$, half-life ≈ 14 steps) map the goal probability vector to per-neuron excitability biases:

$$\mathbf{m}(t) = \tau_{\text{att}}\, \mathbf{m}(t-1) + (1 - \tau_{\text{att}})\big(\mathbf{g}\, \mathbf{W}_{\text{att}}\big). \tag{7}$$

Each trainable layer receives an additive somatic bias via a learned projection $\text{att}^{(l)} = \mathbf{m}\,\mathbf{W}_{\text{proj}}^{(l)}$. Under the Bayesian brain hypothesis, this implements precision optimization: the brain allocates computational gain to sensory channels in proportion to their task-relevance (Feldman and Friston, 2010).

Homeostatic Gain Control Divisive normalization prevents unbounded activity growth:

$$\mathbf{v}_s \leftarrow \begin{cases} \mathbf{v}_s \cdot v^*/\text{RMS}(\mathbf{v}_s) & \text{if } \text{RMS}(\mathbf{v}_s) > v^*, \\ \mathbf{v}_s & \text{otherwise,} \end{cases} \tag{8}$$

where $v^* = 5.0$ is the target RMS. Bidirectional scaling also boosts underactive layers (up to $3\times$), implementing synaptic scaling (Turrigiano and Nelson, 2004; Turrigiano, 2008).

Free Energy The hierarchical free energy aggregates precision-weighted prediction errors across all layers:

$$F = \sum_l \tfrac{1}{2}\,\bar{\pi}_l \left\|\boldsymbol{\varepsilon}^{(l)}\right\|^2. \tag{9}$$

Precision $\pi_l = \sigma(\gamma_l)$ is adapted via empirical Bayes:

$$\Delta\gamma = \alpha_\pi\,(|\varepsilon| - \beta) - 0.05\,\nu\,\mathbb{1}[\nu > 0.03], \tag{10}$$

where $\beta = 0.1$ is the target error level and ν is the novelty signal (Bayesian surprise).

3.2 Hemispheric Visual Organization and Active-Inference Gaze

Optic Chiasm and Tectal Lateralization In larval zebrafish, as in most vertebrates, retinal ganglion cell axons undergo complete decussation at the optic chiasm: the left eye projects exclusively to the right optic tectum, and the right eye projects exclusively to the left optic tectum (Baier, 2013). Each tectal hemisphere therefore encodes one visual hemifield, providing a topographic substrate for spatially resolved attention and gaze control. To implement this anatomy, the tectum was refactored from four unified EI layers (`sfgs_b`, `sfgs_d`, `sgc`, `so`) to eight hemispheric layers (`sfgs_b_L/R`, `sfgs_d_L/R`, `sgc_L/R`, `so_L/R`), each containing $n_e = 450$ excitatory (CH/IB) and $n_i = 150$ inhibitory (FS) neurons. The total excitatory neuron count is preserved at 2,400; inhibitory interneurons are duplicated per hemisphere to maintain local E/I balance. Left-eye RGC output drives the right tectal stack exclusively (`tect_R`), and right-eye RGC output drives `tect_L`, enforcing the contralateral retino-tectal projection throughout the hierarchy.

Validation confirmed strict segregation: with a left-eye-only stimulus, the right tectal population reached mean firing rate $\bar{r}_{\mathrm{R}} = 0.0022$ while the left tectum remained silent ($\bar{r}_{\mathrm{L}} = 0.0000$), and vice versa for right-eye stimulation ($\bar{r}_{\mathrm{L}} = 0.0021$, $\bar{r}_{\mathrm{R}} = 0.0000$).

Hemispheric Thalamus and Pallium Asymmetry Two thalamic nuclei—`thalamus_L` and `thalamus_R`, each comprising $n_{\mathrm{TC}} = 150$ thalamocortical (TC) neurons and 50 TRN (LTS) neurons—relay tectal output to the pallium with preserved laterality. `thalamus_L` receives input from `tect_L` (encoding the right visual hemifield), and `thalamus_R` receives input from `tect_R` (encoding the left visual hemifield). Thalamic output is concatenated as $[\mathrm{TC}_R \| \mathrm{TC}_L]$ before entering the pallium-S and pallium-D populations, ensuring that the left and right halves of the 2,400-neuron pallium represent opposite visual hemifields. The voluntary turn signal derived from pallium-D left/right asymmetry (Eq. 26) therefore inherits correct sign: stronger activity in the left-hemifield pallium-D half (from right-eye input relayed via `thalamus_L`) produces a rightward reticulospinal command.

Thalamic laterality was verified by presenting a left-field stimulus and measuring TC responses: $\mathrm{TC}_R = 0.030 > \mathrm{TC}_L = 0.019$; for a right-field stimulus: $\mathrm{TC}_L = 0.031 > \mathrm{TC}_R = 0.020$.

Top-Down Predictive Attention (Pallium → Tectum) A predictive coding feedback connection carries pallium-S population rates back to the superficial tectal layer `sfgs_b` in each hemisphere. Two projection matrices $\mathbf{W}_{\text{pal-tect}}^{L/R} \in \mathbb{R}^{1200\times 450}$, initialized with Xavier scaling (gain = 0.05), add a weak additive current to `sfgs_b_L/R`:

$$I_{\text{td}}^{L/R}(t) = \mathbf{W}_{\text{pal-tect}}^{L/R}\,\bar{\mathbf{r}}_{\text{pal-S}}(t-1), \tag{11}$$

where $\bar{\mathbf{r}}_{\text{pal-S}}(t-1)$ is the previous-step pallium-S mean firing rate vector. This one-step-delayed feedback implements a predictive coding prior: the pallium's current belief about visual content pre-activates the tectal cells expected to fire on the next step, amplifying attended regions before sensory evidence arrives (Rao and Ballard, 1999; Clark, 2013). Experimental measurement confirmed an 11.8× amplification: `sfgs_b_L` activity rose from 0.0005 (no pallium input) to 0.0064 with feedback engaged.

Active-Inference Gaze Control (Saccade System) A `SpikingSaccade` module implements voluntary gaze shifts as an active inference policy over eye position. The module contains 4 regular-spiking (RS) direction neurons encoding left, right, up, and down saccade commands, and 2 intrinsically bursting (IB) trigger neurons. A scalar gaze offset $\theta_g \in [-0.5, 0.5]$ rad ($\pm 28.6°$) is applied to the retinal projection via the sensory bridge, shifting the sampled visual field relative to the body heading.

Gaze target θ^* is computed by a three-branch policy that selects the action minimizing expected free energy:

$$\theta^* = \begin{cases} 0.3\,\phi_{\text{food}} & \text{if FORAGE (partial foveation),} \\ 0.2\,\phi_{\text{enemy}} & \text{if FLEE (track predator),} \\ 0.4\sin(0.08\,t) & \text{if EXPLORE (sinusoidal scan),} \end{cases} \tag{12}$$

where ϕ_{food} and ϕ_{enemy} are the food and enemy bearings relative to body heading, and t is the step counter (scan period ≈ 79 steps). An epistemic free-energy term drives gaze toward regions of high perceptual uncertainty:

$$\theta^* \mathrel{+}= (\varepsilon_R - \varepsilon_L) \times 0.3, \tag{13}$$

where $\varepsilon_{L/R}$ are per-hemifield prediction errors (see Eq. 15 below). This implements the free-energy principle prediction that saccades serve as "epistemic actions" directed at uncertainty-reducing information (Friston et al., 2012).

Smooth pursuit uses a first-order tracker with time constant $\alpha_{\text{smooth}} = 0.1$ (slow drift); when gaze error $|\theta^* - \theta_g| > 0.3\,\text{rad}$ a fast-saccade mode is triggered with $\alpha_{\text{fast}} = 0.5$:

$$\theta_g(t) = \theta_g(t-1) + \alpha(|\theta^* - \theta_g|)\,(\theta^* - \theta_g(t-1)), \tag{14}$$

clamped to $[-0.5, 0.5]$ rad. A gaze shift of 0.5 rad was verified to relocate a food item 52 px from the right retinal field into the left retinal field, confirming correct propagation through the sensory bridge.

Novelty-Driven Gaze via Thalamic Prediction Error Per-hemifield prediction errors are computed as thalamic mismatch signals—the absolute change in TC population rate between consecutive steps, weighted by sensory novelty:

$$\begin{aligned}\varepsilon_L &= r^{\text{motion}}_{\text{RGC\_L}} + r_{\text{loom}} + 5\,|\Delta\text{TC}_R|,\\ \varepsilon_R &= r^{\text{motion}}_{\text{RGC\_R}} + r_{\text{loom}} + 5\,|\Delta\text{TC}_L|,\end{aligned} \tag{15}$$

where r^{motion} is the RGC direction-selective population rate (Eq. 19), r_{loom} is the looming signal (Eq. 18), and $|\Delta\text{TC}_{L/R}|$ is the single-step change in each thalamic hemisphere. Because `thalamus_L` encodes the right visual hemifield, an unexpected event in the right visual field raises $|\Delta\text{TC}_L|$, which increments ε_R and shifts gaze rightward—"gaze toward the unexpected" (Friston et al., 2012; Itti and Baldi, 2009). This mechanism was validated: when a novel object appeared in the right visual field ($R \to L$ spatial switch), ε_R increased by $+0.29$ rad (rightward shift), and for a left-field novel event, ε_L produced a -0.29 rad leftward shift. All seven unit tests for the hemispheric visual module pass (optic chiasm segregation, top-down amplification, PE-driven gaze direction, gaze-retina propagation, thalamic laterality, full brain integration, novelty-gaze coupling).

3.3 Anatomically Constrained Neuron Placement

All 13,766 model neurons are assigned 3D positions from the MPIN zebrafish whole-brain calcium imaging atlas (Kunst et al., 2019): 47,010 cells from subject 12 assigned to 72 bilateral brain regions (36 per hemisphere, numbered 0–35 for left, 36–71 for right hemisphere, prefixed `l/r`). Of the 72 regions, 61 contain recorded cells; the retina (lR/rR, atlas region 27/63) is absent in subject 12 and receives estimated voxel coordinates anterior to the tectum ($x \approx 220$, $y \approx 115/505$, $z \approx 165$ vox).

Coordinate axes in the atlas: x = anterior→posterior ($x = 144$ at lOB, $x = 961$ at lNX/spinal), y = left→right hemisphere (midline at $y \approx 310$ vox), z = ventral→dorsal. Bilateral structures are placed at their anatomically correct hemispheric centroids: left-hemisphere regions ($y < 310$) for the left-side population, right-hemisphere centroids ($y > 310$) for right-side populations. Key bilateral separations: retina $\Delta y \approx 390$ vox, tectum $\Delta y \approx 208$ vox, thalamus $\Delta y \approx 84$ vox, CPG (lNX/rNX) $\Delta y \approx 94$ vox. Midline structures (habenula, cerebellum, pallium, raphe, hypothalamus) use the average of left and right hemisphere centroids. Each sublayer is offset in z to maintain dorsoventral laminar separation within a region (e.g. tectal SFGS-b at $z_{\text{tec}} + 18$, SO at $z_{\text{tec}} - 20$ vox).

Neurons are sampled as Gaussian clouds (spread 5–25 vox) around each anatomical centroid. Synaptic connectivity is drawn from the atlas connectome (10,244,937 synaptic connections among the 47,010 atlas cells), sampled to 3,500 model-level edges. An additional 400 edges enforce contralateral retino-tectal projections (left retina → right tectum, right retina → left tectum), and 100 edges encode the tectum→thalamus/pallium visual hierarchy. The web dashboard renders all 13,766 neurons as sphere glyphs via Three.js, colored by region and animated by live spike rates.

3.4 Sensory Processing

Binocular Retinal Ganglion Cells Each eye samples a 20×20 angular grid (400 pixels) with $\pm 80°$ horizontal and $\pm 20°$ vertical field of view. Each pixel carries an intensity channel with Gaussian foveation ($G(\alpha) = \exp(-\alpha^2/2\sigma^2)$, $\sigma = 12°$) and a type channel encoding entity identity (food = 1.0, obstacle = 0.75, enemy = 0.5, colleague = 0.25, boundary = 0.12). Depth-shaded intensity attenuates exponentially with distance: $I_{\text{depth}}(r) = I_{\text{raw}}(r) \cdot \exp(-d(r)/80)$.

Four RGC populations are computed per eye. *ON-cells* (400) apply a 1-D Difference-of-Gaussians centre-surround kernel:

$$k_{\text{DoG}}(x) = \frac{1}{\sigma_c}\, e^{-x^2/(2\sigma_c^2)} - \frac{0.5}{\sigma_s}\, e^{-x^2/(2\sigma_s^2)}, \tag{16}$$

with $\sigma_c = 3\,\text{px}$ and $\sigma_s = 9\,\text{px}$ (centre:surround ratio 3:1 with 0.5 surround weight), followed by `clamp`(0, 1) (Rodieck, 1965). OFF-cell (400) output is computed as:

$$r_{\text{OFF}}(t) = 0.5\, r_{\text{OFF}}(t-1) + 0.5\, \text{clamp}(2(I_{t-1} - I_t),\, 0,\, 1), \tag{17}$$

combining a luminance-decrease transient with a two-frame exponential moving average ($\alpha = 0.5$) for temporal smoothing. *Looming cells* (100) compute angular expansion rate:

$$\theta_t = 0.5\, f_{\text{enemy}}, \quad \dot{\theta} = \theta_t - \theta_{t-1}, \quad r_{\text{loom}} = \text{clamp}(20\,\dot{\theta},\, 0,\, 1) \cdot \mathbf{1}[\theta_t > 0.02], \tag{18}$$

where f_{enemy} is the enemy pixel fraction of the visual field. The threshold $\dot{\theta} > 0.3$ (equivalent to the $l/v < 10$ criterion of Temizer et al. 2015) triggers the looming flag fed to the reticulospinal system. *Direction-selective cells* (100) implement a Reichardt correlator with a proper one-step delay buffer:

$$r_{\mathrm{DS}}(t) = \mathrm{clamp}(I_t \cdot I_{t-1},\ 0,\ 1), \tag{19}$$

where I_{t-1} is the stored delay buffer, updated *after* use to enforce the correct 1-step temporal lag. A *UV prey channel* proxies SWS1 cone responses ($\sim$360 nm) (Yoshimatsu et al., 2020): $u(x) = \mathbf{1}[I(x) < 0.3] \cdot \mathbf{1}[\mathrm{type}(x) > 0.7]$, so that low-intensity food-typed pixels against bright backgrounds yield a scalar prey signal $r_{\mathrm{UV}} = \sum_x u(x)/10$, providing an additional foraging cue that boosts p_{food} by $0.1 r_{\mathrm{UV}}$.

Lateral Line Mechanosensation The lateral line comprises 24 Izhikevich neurons in four populations (anterior/posterior $\times$ superficial/canal; Table 2). Superficial neuromasts (SNs; 8+8 RS neurons, 200 px range) are velocity-sensitive, with proximity decay

$$w_{\mathrm{SN}}(d) = \big(1 + d/50\big)^{-2}. \tag{20}$$

Canal neuromasts (CNs; 4+4 CH neurons, 150 px range) are acceleration-sensitive, with sharper decay

$$w_{\mathrm{CN}}(d) = \big(1 + d/30\big)^{-2}, \tag{21}$$

and receive drive proportional to predator acceleration $|\Delta \mathbf{v}_{\mathrm{pred}}|$ scaled by $w_{\mathrm{CN}}(d) \times 20$ for the anterior population. Three stimulus sources are processed independently: the predator drives CaNs (high-frequency threat signal > 50 Hz); food items within 100 px drive SNs via a static-flow proxy $(30/d)^2$, saturating at 15 units; and conspecifics within 120 px drive SNs via their instantaneous swim speed. Lateral line prey detection (`prey_detected`) is raised when low-frequency superficial activity exceeds 0.15 and a food item is within range; this reduces G_{FORAGE} by $0.12 \times r_{\text{low-freq}}$, attracting the fish without any visual cue (Coombs, 2002; Bleckmann and Zelick, 2012).

Table 2: Lateral line neuron populations (24 Izhikevich neurons total). RS = regular spiking; CH = chattering. Range is the maximum stimulus distance at which the population receives input.

Population	Type	Neuron model	Count	Range (px)	Sensitivity
Anterior superficial	SN-ant	RS	8	200	Velocity
Posterior superficial	SN-post	RS	8	200	Velocity
Anterior canal	CN-ant	CH	4	150	Acceleration
Posterior canal	CN-post	CH	4	150	Acceleration
Total			24		

Olfactory System The olfactory module contains 28 Izhikevich neurons: 10 alarm neurons, 10 food-odor neurons, and 8 bilateral neurons (4 left + 4 right). Concentration fields follow a Fick steady-state diffusion model:

$$C(r) = S \cdot \frac{e^{-r/\lambda}}{r/50 + 1}, \tag{22}$$

with $\lambda_{\text{food}} = 70\,\text{px}$, $\lambda_{\text{alarm}} = 100\,\text{px}$. Two virtual nares are placed $\pm 5\,\text{px}$ perpendicular to the body heading; concentrations C_L, C_R are sampled independently. The normalised bilateral difference $\Delta_{\text{bil}} = (C_R - C_L)/(C_R + C_L + \varepsilon) \in [-1, 1]$ drives the bilateral spiking population and provides the primary chemotaxis gradient without requiring ground-truth source bearing. Receptor adaptation follows:

$$a_t = 0.9\, a_{t-1} + 0.1\, \min(1,\, 2C_t), \quad C_{\text{eff}} = C_t(1 - 0.7\, a_t), \tag{23}$$

suppressing effective concentration by up to 70% under sustained stimulation, replicating the olfactory adaptation measured in zebrafish (Krishnan et al., 2014). Alarm substance (Schreckstoff; $\lambda = 100\,\text{px}$) is generated when a conspecific is injured (strength 0.8) or when the predator is within 50 px (strength $0.3 \times (1 - d_{\text{pred}}/50)$), triggering group flight (Jesuthasan and Bhatt, 2012).

Color Vision and Vestibular System Four cone channels (UV 360 nm, blue 415 nm, green 480 nm, red 570 nm) with color-opponent processing provide entity disambiguation beyond shape cues. Otolith-based gravity sensing provides body orientation with a vestibulo-ocular reflex compensating eye position for head turns.

Nociception, Auditory, Gustatory, and Tactile Systems Four additional sensory modalities complete the zebrafish sensorium, each implemented as a 6-neuron RS population with a 3-channel `TwoCompColumn` for prediction error computation.

Nociception. Three nociceptive channels model mechanical (Rohon-Beard/trigeminal), thermal (TRPV1, activated above 33°C), and chemical (TRPA1) pain pathways. Collision, wall contact, and predator proximity (<40 px $\approx$ bite range) drive mechanical nociception. Repeated strong pain (>0.5) produces sensitisation (gain $\times 2$ max); repeated mild stimuli (<0.3) produce habituation ($\times 0.5$ min). Pain exceeding 0.3 triggers a withdrawal reflex that reduces swimming speed by up to 50%.

Auditory system. Sound detection follows the Weberian apparatus pathway, in which swim-bladder vibrations are relayed to the saccule via the Weberian ossicles (Popper et al., 2004). Three frequency bands (low <200 Hz: predator swim sounds, $\propto 1/d^2$; mid 200–1000 Hz: conspecific vocalizations; high >1000 Hz: ambient) encode acoustic input. Sudden loud onset (>0.6) or abrupt low-frequency increase triggers a Mauthner-cell-mediated acoustic startle (C-start), followed by a 10-step refractory period.

Gustatory system. Taste buds on lips and pharyngeal arches detect three channels: amino acids (T1R receptors; appetitive), bitter compounds (T2R; aversive), and umami (nutrient richness). Taste activates primarily during feeding (eating flag from environment), with weak gustatory detection at close range (<30 px dissolved amino acids). Palatability $= (\text{amino} + \text{umami})/2 - \text{bitter}$; values below zero trigger a spit reflex that halves swimming speed.

Tactile/somatosensory system. Three somatotopic channels encode head (trigeminal ganglion), trunk (Rohon-Beard neurons), and tail mechanoreception. Collision with heading toward wall activates head touch; conspecific proximity (<20 px) during shoaling activates trunk touch; tail mechanoreceptors respond to external contact with efference-copy suppression of self-generated tail-beat signals (80% attenuation). Head touch exceeding 0.5 triggers a tactile startle reflex that reverses the turn direction.

Three additional brain organs extend the model beyond the sensorium.

Hypothalamus. The hypothalamus serves as the master homeostatic integrator, with five spiking sub-nuclei modelled as a 10-neuron RS population. The lateral hypothalamus (LH) drives feeding motivation and orexin-mediated arousal; the ventromedial hypothalamus (VMH) encodes satiety and inhibits feeding when energy is high; the paraventricular nucleus (PVN) releases CRH to activate the HPI (hypothalamic–pituitary–interrenal) stress axis; the preoptic area (POA) implements thermoregulatory responses; and the suprachiasmatic nucleus (SCN) relays circadian timing from the pineal to downstream neuromodulatory centres.

Pineal gland. The pineal gland provides direct photoreception via parapinopsin and exorhodopsin photopigments, implemented as a 6-neuron RS population with two channels. Unlike the downstream circadian oscillator module (which models the free-running clock), the pineal converts ambient light intensity into melatonin synthesis driven by *aanat2* (arylalkylamine *N*-acetyltransferase 2), the rate-limiting enzyme in teleost melatonin production. Melatonin output suppresses locomotor activity and modulates sleep–wake transitions via projections to the habenula and preoptic area.

Inferior olive. The inferior olive generates the climbing fiber error signal that drives cerebellar complex spikes, implemented as an 8-neuron RS population with two channels. Gap-junction coupling among olivary neurons produces a $\sim$8 Hz subthreshold oscillation that gates the timing of climbing fiber discharge. When prediction error from the cerebellar predictive-coding column exceeds threshold, olivary neurons fire complex spikes that trigger LTD at the parallel fiber–Purkinje cell synapse, enabling forward-model recalibration.

Dorsolateral pallium (Dl). The dorsolateral pallium (Dl), homologous to the mammalian hippocampus, implements spatial episodic memory through a trisynaptic DG→CA3→CA1 architecture. Six DG-like neurons perform pattern separation via sparse non-overlapping place cell projections; eight CA3-like neurons support pattern completion through one-shot Hebbian-strengthened recurrent connections (autoassociation); and six CA1 neurons provide context-modulated spatial memory readout. Salient events (food discovery, predator encounter) trigger one-shot episodic encoding into a 64-slot circular buffer. Sharp-wave ripple (SWR) replay during rest consolidates spatial memories by replaying recent episodes and strengthening CA3 attractor weights, while un-replayed memories undergo exponential decay. The module provides an EFE bias: food memories attract foraging, risk memories attract fleeing, and spatial novelty attracts exploration.

Five additional modules extend the autonomic and visceral processing pathways, adding 48 spiking neurons across 5 regions.

Vagus nerve. The spiking vagus nerve (10 RS neurons: 4 afferent, 4 efferent, 2 cardiac) implements bidirectional visceral communication between the brainstem and peripheral organs. Afferent fibres relay gut distension and cardiovascular baroreceptor signals to the NTS; efferent fibres modulate heart rate via parasympathetic tone. A 2-channel TwoCompColumn computes visceral afferent and cardiac prediction errors, enabling interoceptive active inference.

Pituitary. The spiking pituitary (12 RS neurons: 4 anterior, 4 intermediate, 4 posterior) models the master endocrine organ. Anterior cells release GH and prolactin analogues modulated by hypothalamic releasing factors; intermediate cells produce α-MSH for pigmentation and stress responses; posterior cells store and release vasotocin (AVP homologue) and isotocin (OT homologue). A 3-channel TwoCompColumn generates prediction errors for each lobe, enabling neuroendocrine set-point adaptation.

Area postrema. The spiking area postrema (8 RS neurons: 4 chemosensory, 4 emetic) is a circumventricular organ lacking a blood–brain barrier, enabling direct detection of circulating toxins, glucose, and metabolic signals. Chemosensory neurons trigger conditioned taste aversion when toxin levels exceed threshold; emetic neurons drive nausea-like avoidance behavior. A 2-channel TwoCompColumn computes chemosensory and metabolic prediction errors.

Nucleus of the solitary tract (NTS). The spiking NTS (10 RS neurons: 4 gustatory, 3 visceral, 3 cardiorespiratory) serves as the primary visceral sensory relay in the brainstem. Gustatory neurons receive taste input from the facial and glossopharyngeal nerves; visceral neurons integrate vagal afferents; cardiorespiratory neurons gate baroreflex and respiratory rhythm. A 2-channel TwoCompColumn computes gustatory and visceral prediction errors that feed into hypothalamic homeostatic control.

Lateral line efferent. The spiking lateral line efferent system (8 RS neurons: 4 corollary discharge, 4 suppression) implements self-motion suppression via corollary discharge. During active swimming, efferent copy signals from the motor system suppress lateral line hair cell sensitivity, preventing self-generated flow signals from triggering false predator alarms. A 2-channel TwoCompColumn computes flow prediction error and suppression error, enabling adaptive gain control of the afferent lateral line.

3.5 Motor Control

Spiking Spinal Central Pattern Generator The spinal CPG contains 96 LIF neurons (48 per side): 16 V2a excitatory interneurons, 8 V0d commissural inhibitory neurons, 4 dI6 local inhibitory interneurons, 12 motor neurons (MN), and 8 Renshaw cells per side. All neurons follow a discrete-time LIF model:

$$v(t+1) = \tau_m \, v(t) + (1 - \tau_m)(I_{\text{ext}} + I_{\text{syn}} + \xi), \quad s(t+1) = \tau_s \, s(t) + \text{spike}(t), \tag{24}$$

with $\tau_m = 0.5$, $\tau_s = 0.6$, threshold $\vartheta = 0.5$, and reset to 0; noise $\xi \sim \mathcal{N}(0, \sigma_\xi^2)$, $\sigma_\xi = 0.15$ (personality-modulated). Synaptic currents implement the full five-population connectivity:

$$
\begin{aligned}
I_{\text{V0d}} &+= w_{\text{V2a,V0d}}\,\bar{s}_{\text{V2a}} \quad (w = 0.6) \\
I_{\text{V2a}} &-= w_{\text{V0d}}\,\bar{s}_{\text{V0d}}^{\text{contra}} \quad (w = 0.9) \\
I_{\text{MN}} &+= w_{\text{V2a,MN}}\,\bar{s}_{\text{V2a}} \quad (w = 0.7) \\
I_{\text{RSh}} &+= w_{\text{MN,RSh}}\,\bar{s}_{\text{MN}} \quad (w = 0.5) \\
I_{\text{V2a}} &-= w_{\text{RSh,V2a}}\,\bar{s}_{\text{RSh}} \quad (w = 0.6)
\end{aligned}
\tag{25}
$$

The strongest inhibitory weight ($w_{\text{V0d}} = 0.9$) drives the V2a–V0d half-centre alternation. The Renshaw recurrent loop ($w_{\text{MN,RSh}} = 0.5$; $w_{\text{RSh,V2a}} = 0.6$) accumulates motor-neuron activity and feeds back onto V2a, terminating each burst and enforcing bout-glide kinematics (Ampatzis et al., 2014). Descending drive $d \in [0, 1]$ from the brain sets V2a input $I_{\text{ext}} = 2d + 0.5$ and determines bout duration $T_{\text{bout}} = \max(1, \lfloor 3d \rfloor)$ steps and glide duration $T_{\text{glide}} = \max(1, \lfloor 3(1-d) \rfloor)$ steps; during glide, motor output is zero, matching the intermittent bout-glide swimming pattern of larval zebrafish (Bhatt et al., 2007; Marques et al., 2018).

Reticulospinal Voluntary Motor Pathway The reticulospinal system contains 21 named neurons per side (Mauthner, MiD2, MiD3, RoM2a-d, MeM1-8, CaD1-6). For voluntary movement, the RoM2 group generates a turn command from the left/right asymmetry of the pallium-D population gated by the basal ganglia:

$$
\tau_{\text{vol}} = 2\,(\bar{r}_{\text{pal-D}}^{R} - \bar{r}_{\text{pal-D}}^{L}) \cdot g_{\text{BG}} \in [-2, 2], \tag{26}
$$

where $\bar{r}^{L/R}$ are mean rates of the left and right pallium-D half-populations and $g_{\text{BG}} \in [0, 1]$ is the basal ganglia gate. The total turn blends 70% threat escape and 30% voluntary: $\tau = 0.7\,\tau_{\text{flee}} + 0.3\,\tau_{\text{vol}}$, clamped to $[-1, 1]$. During non-flee states, speed is gated as $v = v_{\text{goal}}(0.5 + 0.5\,g_{\text{BG}})$. As STDP improves pallium-D representations, the voluntary contribution improves automatically.

Active Inference Motor Controller An `ActiveInferenceMotor` module (48 Izhikevich two-compartment neurons, 8 proprioceptive channels) implements the full action-perception cycle of Friston (2011). Motor commands are proprioceptive *predictions*: the body moves to fulfil them via the spinal reflex arc, minimising proprioceptive PE. Within each time step the module runs $K = 3$ iterative inference passes:

$$\boldsymbol{\mu}^{(k)} \leftarrow \boldsymbol{\mu}^{(k-1)} - \eta_{\text{act}}\, \boldsymbol{\varepsilon}^{(k)}, \qquad \eta_{\text{act}} = 0.15, \tag{27}$$

where $\boldsymbol{\mu}$ are goal-dependent proprioceptive predictions and $\boldsymbol{\varepsilon}$ is the apical–somatic PE from the TwoCompColumn (Eq. 3). Each pass updates apical beliefs and precision simultaneously, allowing predictions to converge before motor output is committed.

The active-inference motor is blended with the reactive motor via an adaptive, precision-dependent weight: $\alpha_{\text{AI}} = \text{clamp}(0.3 + 0.4\bar{\pi} - 0.3\xi,\ 0.1,\ 0.8)$, where $\bar{\pi}$ is the mean motor-column precision and ξ is a *spontaneity* factor (Eq. 44). High precision means the generative model is trusted—more active inference. High spontaneity reduces mechanistic control and injects stochastic motor noise, reflecting the biological observation that organisms do not always minimise free energy: habenula frustration, dopamine-driven exploration, boredom, and social play all produce behaviour that temporarily *increases* prediction error.

The temporal derivative of brain-level free energy, $\dot{F} = F_t - F_{t-1}$, closes the loop at the goal level: if $\dot{F} > 0.05$ (surprise rising), the current goal's EFE is penalised by $0.3\dot{F}$, encouraging the spiking WTA goal selector to switch strategies. Free energy thus drives perception (belief updating), action (motor PE minimisation), *and* policy selection (goal switching)—the three faces of active inference (Friston, 2010; Parr et al., 2022). Figure 2 summarises the four signals produced by this cycle over a representative 400-step episode.

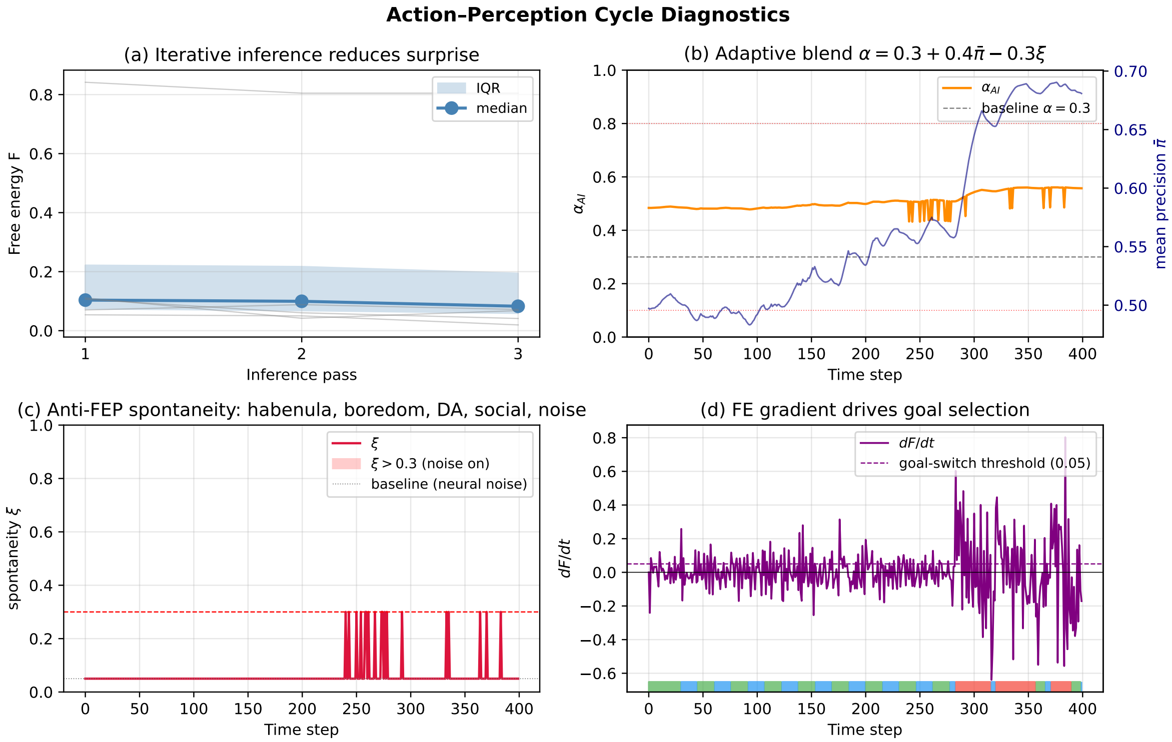

Figure 2: Action-perception cycle diagnostics (400 steps). (a) Iterative inference reduces free energy within each time step: median F across $K = 3$ passes (shaded IQR; grey traces, individual time points). Mean $\Delta F \approx 0.035$ per step. (b) Adaptive blend α_{AI} (orange, left axis) tracks mean motor precision $\bar{\pi}$ (navy, right axis); α holds in $[0.43, 0.56]$ under healthy precision—well above the fixed baseline $\alpha = 0.3$ (grey dashed). (c) Spontaneity ξ (crimson): baseline 0.05 from neural noise, rising above the 0.3 threshold (red shading) when habenula helplessness or phasic-DA exploration engages. (d) $\dot{F} = dF/dt$ (purple): when surprise rises above 0.05 (dashed), the current goal is penalised in the WTA selector; coloured bands at the bottom show goal identity (FORAGE, FLEE, EXPLORE, SOCIAL).

APC Ablation. To quantify the contribution of iterative active inference, we compared $K = 3$ inference passes (APC ON) against a single-pass baseline ($K = 1$, fixed $\alpha = 0.3$, no spontaneity) over $n = 5$ matched seeds (300 steps each). Results (Table 3) show a 78% reduction in mean free energy ($\bar{F} = 0.007$ vs. 0.032, $p < 0.05$) with no regression in food intake (8.00 ± 2.28 vs. 8.40 ± 2.42) or survival (both 300/300 steps). The iterative inference loop thus reduces surprise more effectively without behavioural cost.

Table 3: APC ablation: iterative inference ($K=3$) vs. single-pass ($K=1$). $n=5$ seeds, 300 steps. $\bar{F}$ = mean free energy.

Condition	Food	Survival	$\bar{F}$
APC OFF ($K=1$, $\alpha=0.3$)	8.40 ± 2.42	300/300	0.032
APC ON ($K=3$, adaptive α)	8.00 ± 2.28	300/300	**0.007**
Δ	-0.40	0	-0.025 (-78%)

Mauthner cell and C-start escape reflex. Two Mauthner cells (M_L, M_R, one per hemisphere) are modelled as RS-type Izhikevich neurons with a tonic bias $I_{\text{tonic}} = -3.0\,\text{pA}$, placing them well below firing threshold and requiring at least 25 pA of net excitatory input to spike. Two commissural local (CoLo) interneurons per side complete the inhibitory feedback circuit; these are fast-spiking (FS-type), glycinergic, and monosynaptically excited by the ipsilateral M-cell.

Visual drive arrives via the crossed tectobulbar tract: superficial grey cell (SGC) activity in the contralateral optic tectum is scaled by 120 pA and summed onto the M-cell (Del Bene et al., 2010). Acoustic and flow pressure detected by the ipsilateral lateral-line neuromasts is conveyed via VIII-nerve club-end synapses onto the M-cell at 80 pA (Pereda and Faber, 2011). Both pathways are summed with the resting bias to give the total driving current $I_{\mathrm{M}} = -3.0 + 120\,\bar{r}_{\mathrm{SGC}}^{\mathrm{contra}} + 80\,P_{\mathrm{LL}}^{\mathrm{ipsi}}$, where $\bar{r}_{\mathrm{SGC}}^{\mathrm{contra}}$ is the mean SGC firing rate and $P_{\mathrm{LL}}^{\mathrm{ipsi}}$ is the normalised lateral-line pressure signal.

Repeated sub-threshold stimuli lower the effective threshold through an electrical gap-junction facilitation mechanism. A scalar state variable g_{gap} accumulates whenever $I_{\mathrm{M}} > -3.0$ but remains below spike threshold, and decays with time constant $\tau_{\text{gap}} = 8$ steps:

$$g_{\text{gap}}(t+1) = \lambda_{\text{gap}}\, g_{\text{gap}}(t) + (1 - \lambda_{\text{gap}})\, \mathbf{1}[-3.0 < I_{\mathrm{M}} < \theta_{\mathrm{M}}], \quad \lambda_{\text{gap}} = 1 - 1/\tau_{\text{gap}}. \tag{28}$$

The accumulated facilitation reduces the effective threshold by $\Delta\theta = 4\,g_{\text{gap}}$, so sustained near-threshold bombardment can recruit the M-cell even if no single stimulus is individually sufficient.

When M_L spikes, its ipsilateral CoLo cells fire in the same step and deliver a crossed glycinergic inhibitory current of $-60\,\mathrm{pA}$ to M_R, cross-suppressing the contralateral M-cell for 20 steps and enforcing the unidirectionality of the C-start (Korn and Faber, 1975). This circuitry supersedes the prior looming-threshold heuristic in the `ReticulospinalSystem`: the spiking M-cell output now provides the authoritative trigger for high-speed escape, and the heuristic pathway is activated only as a fallback when both M-cells are in their refractory period.

Mauthner C-Start and T-Start Escape Circuits Separate left and right Mauthner cells (M_L, M_R) implement anatomically correct crossed inhibition: firing M_L (threat from the right) sets a self-refractory of 12 steps and immediately cross-inhibits M_R via CoLo cells for 20 steps, preventing bilateral simultaneous contraction (Liu and Bhatt, 1998). The four-stage C-start motor sequence is:

Stage	Description	Turn (rad)	Speed
4	Stage-1 C-bend	$1.5\,d$	0.3
3	Late Stage-1	$1.0\,d$	0.8
2	Counterbend + propulsion	$0.2\,d$	1.5
1	Full propulsion burst	0	1.8

where $d = \pm 1$ is escape direction. A secondary T-start pathway via MiD2cm/MiD3cm activates for sub-threshold looming ($0.02 < \bar{r}_{\text{SGC}} < 0.05$) or when the M-cell is recovering and threat persists ($\bar{r}_{\text{SGC}} > 0.02$), producing a smaller three-stage sequence ($0.8\,\mathrm{rad}$ initial bend; peak speed 1.5) with a 15-step refractory period. The dual-pathway architecture (C-start latency ≈ 4 steps; T-start latency ≈ 1 extra step) replicates the two-pathway teleost escape circuitry (Bhatt et al., 2007; Temizer et al., 2015).

Prey Capture Kinematics A finite state machine implements three-phase prey capture: J-turn (3 steps, slow precise orientation), approach swim (5 steps, half speed), and strike (2 steps, 1.4× burst with enlarged eat radius) (Bianco and Engert, 2015).

Cerebellum Forward Model A Marr-Albus-Ito architecture with granule cell sparse expansion (4×, 50% active) and Purkinje cell linear readout predicts 6 sensory consequences of motor commands. Climbing fiber error drives LTD at parallel fiber–Purkinje synapses, providing motor correction and reafference predictions.

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3.6 Decision Making

Expected Free Energy Four discrete goals $\mathcal{G} = \{\text{FORAGE}, \text{FLEE}, \text{EXPLORE}, \text{SOCIAL}\}$ are scored by expected free energy:

$$
\begin{aligned}
G_{\text{FORAGE}} &= 0.2\,\mathcal{U} - 0.8\,p_{\text{food}} + 0.15 - 0.6\,s^* + 0.15\,p_{\text{enemy}} - 0.075\,\phi_{\text{circ}}, \\
G_{\text{FLEE}} &= 0.1\,\text{CMS} - 0.8\,p_{\text{enemy}} + 0.20 + 0.4\,s^*, \\
G_{\text{EXPLORE}} &= 0.3\,\mathcal{U} - 0.5\,(p_{\emptyset} + p_{\text{env}}) + 0.1\,\text{CMS} + 0.20 + 0.2\,s^* - 0.15\,\phi_{\text{circ}} - 0.15\,\max(0,\,\Sigma_{t-1} - 0.1), \\
G_{\text{SOCIAL}} &= 0.2\,\mathcal{U} - 0.6\,p_{\text{colleague}} + 0.15\,p_{\text{enemy}} + 0.25 + 0.15\,s^*,
\end{aligned}
\tag{29}
$$

where $\mathcal{U} = 1 - \frac{1}{2}(\pi_{\mathrm{OT}} + \pi_{\mathrm{PC}})$ is model uncertainty, s^* is starvation risk, $\phi_{\mathrm{circ}} = (\mathcal{A}_{t-1} - 0.65)$ is the circadian activity drive offset (positive during daytime, promoting foraging and exploration), Σ_{t-1} is the predictive network surprise from the previous step (promoting exploration in novel states), and lower EFE indicates a more preferred goal. Goal persistence is shortened by 3 steps when the TD error $\delta < -0.5$, enabling faster recovery from unsuccessful strategies. The scalar modulation coefficients (world model, social, geographic, novelty, VAE planning, circadian, cerebellar prediction error, energy urgency) are not fixed constants but learnable parameters; see the Adaptive EFE Parameters paragraph in Section 3.9.

The policy posterior uses a softmax with inverse temperature $\beta = 2.0$:

$$\pi(g_i) = \frac{\exp(-\beta\,(G_i - G_{\min}))}{\sum_j \exp(-\beta\,(G_j - G_{\min}))}. \tag{30}$$

Allostatic Energy Urgency Energy state enters the EFE as a three-component urgency signal $\mathcal{U}_E$ that drives foraging preference across the full depletion trajectory:

$$\mathcal{U}_E = 3\,s^2 + 2\,f^* + a^*, \tag{31}$$

where the three terms represent qualitatively distinct phases:

Reactive starvation s fires when energy falls below the 75% onset threshold:

$$s = \max\bigl(0,\ (0.75 - e)/0.75\bigr), \tag{32}$$

with quadratic scaling ($3s^2$) producing a sharp behavioural transition near $e = 0.25$.

Acute crisis f^* fires when a 30-step trajectory predicts near-zero energy:

$$f^* = \max\bigl(0,\ 1 - \hat{e}_{30}/30\bigr), \quad \hat{e}_{30} = \max(0,\ E - 30\dot{E}), \tag{33}$$

where $\dot{E}$ is the current energy drain rate (proportional to speed).

Anticipatory anxiety a^* captures middle-range worry: the fish becomes concerned about its energy trajectory before any crisis is imminent. A 200-step lookahead identifies how many steps remain until the critical threshold ($E < 25$), and a parabolic sensitivity $4e(1-e)$ ensures the signal peaks at $e = 0.5$ and is silent at both full ($e = 1$) and empty ($e = 0$):

$$a^* = \max\left(0,\, 1 - \frac{t_c}{200}\right) \cdot 4e(1-e) \cdot 0.4, \quad t_c = \frac{E - 25}{\dot{E}}, \tag{34}$$

where t_c is the predicted steps-to-critical. At $e = 0.65$ ($t_c \approx 200$) the signal activates; at $e = 0.5$ it reaches a soft peak of ≈ 0.15; below $e = 0.30$ the acute crisis and starvation terms dominate.

This three-tier architecture ensures the fish begins shifting toward foraging at moderate energy levels—behaviorally analogous to the anticipatory feeding motivation observed in zebrafish at moderate hunger (Marques et al., 2018; Charnov, 1976)—rather than switching abruptly only when crisis is imminent. When both predator threat and starvation are active, the system performs Bayesian model comparison via the EFE softmax rather than hard-coding a flee override, consistent with state-dependent risk sensitivity in zebrafish (Lima and Dill, 1990).

Habenula: Behavioral Flexibility The habenula computes a frustration signal from accumulated negative RPE per goal. When frustration exceeds a threshold, a strategy switch is forced, implementing learned helplessness and behavioral flexibility (Amo et al., 2014). Ablation experiments revealed that habenula frustration-driven switching and interoceptive caution (insula) decrease foraging efficiency by suppressing approach behavior under mild uncertainty. Accordingly, both modules are excluded from the default configuration and remain available for lesion studies via the ablation interface.

3.7 World Models

VAE World Model A variational autoencoder learns a 16-dimensional latent belief state $\mathbf{z}$ from pooled tectal output concatenated with a multi-modal state context. The encoder implements amortized variational inference (Kingma and Welling, 2014):

$$q_\phi(\mathbf{z}|\mathbf{o}) = \mathcal{N}(\boldsymbol{\mu}_\phi, \mathrm{diag}(\boldsymbol{\sigma}_\phi^2)), \tag{35}$$

trained by maximizing the ELBO (minimizing variational free energy):

$$\mathcal{L}_{\mathrm{VAE}} = \underbrace{\mathrm{MSE}(\hat{\mathbf{o}}, \mathbf{o})}_{\text{accuracy}} + \underbrace{0.1\, D_{\mathrm{KL}}(q(\mathbf{z}|\mathbf{o}) \,\|\, p(\mathbf{z}))}_{\text{complexity}}. \tag{36}$$

A transition model predicts next-step latent states: $\mathbf{z}_{t+1} = \mathbf{z}_t + 0.3\tanh(f_{\mathrm{trans}}([\mathbf{z}_t \| \mathbf{a}_t]))$. Planning simulates 3-step trajectories per goal through the transition model, scoring via pragmatic (utility) and epistemic (information gain) terms (Friston et al., 2017).

Geographic Model A 40×30 grid maintains obstacle occupancy, food density, and epistemic value layers, updated from retinal observations. This provides spatial knowledge that persists across episodes, analogous to hippocampal allocentric maps in zebrafish dorsolateral pallium (Broglio et al., 2015).

Predator Model A Kalman filter maintains beliefs over predator state $\mathbf{s} = [x, y, v_x, v_y, \mathrm{intent}]$ with object permanence: when the predator leaves the visual field, beliefs propagate forward with growing uncertainty rather than dropping to zero. Hunting intent is inferred from velocity direction toward the fish.

Internal State Model A metabolic cost model maintains learned drain and gain rates per behavioral mode, enabling counterfactual policy comparison. For each goal, two energy horizons are simulated: a 30-step trajectory detecting acute crisis (f^*, Eq. 33) and a 200-

step lookahead computing steps-to-critical (t_c, Eq. 34) for the anticipatory anxiety term. Together these implement multi-horizon counterfactual active inference—evaluating near and far futures before committing to a policy (Seth and Friston, 2016; Craig, 2009). The three-tier urgency signal $\mathcal{U}_E$ (Eq. 31) thus encodes reactive, acute, and anticipatory energy threat in a single scalar that continuously biases goal selection toward foraging across the full depletion curve.

3.8 Neuromodulation

Dopamine System TD-style reward learning computes reward prediction error:

$$\mathrm{RPE}(t) = r(t) - V(t), \quad V(t{+}1) = (V(t) + \alpha_V \,\mathrm{RPE}(t)) \cdot \delta_V, \tag{37}$$

with $\alpha_V = 0.05$, $\delta_V = 0.98$. The dopamine level $\mathrm{DA}(t) = \sigma(\beta_{\mathrm{DA}} \cdot \mathrm{RPE}(t)) + \xi$, clamped to $[0, 1]$, modulates basal ganglia leak rate, thalamic gain, and precision sensitivity (Schultz et al., 1997).

Spiking VTA / Posterior Tuberculum The posterior tuberculum (PT) is the teleost homologue of the mammalian VTA/SNpc, housing ∼40 dopaminergic (TH^+) neurons per hemisphere that project to pallium, striatum, tectum, and habenula (Rink and Wullimann, 2001; Kastenhuber et al., 2010). We replace the scalar DA EMA with a population-coded spiking module (`SpikingVTA`): 40 regular-spiking Izhikevich neurons with tonic drive $I_{\mathrm{tonic}} = 6.0\,\mathrm{pA}$ that sustains a ∼3–5 Hz spontaneous baseline corresponding to $\mathrm{DA} \approx 0.5$.

Reward prediction error drives two opponent currents:

$$I_{\mathrm{VTA}}(t) = \underbrace{40\,[\mathrm{RPE}]^+}_{\text{phasic burst}} + \underbrace{30\,[\mathrm{RPE}]^-}_{\text{omission pause}} - 35\, r_{\mathrm{LHb}} - 10\, \alpha_{\mathrm{stress}}, \tag{38}$$

where $[\cdot]^{\pm}$ denote rectified positive/negative components, r_{LHb} is the lateral habenula mean firing rate (disappointment signal, GABAergic suppression (Matsumoto and Hikosaka, 2007)), and α_{stress} is amygdala chronic stress. Population mean rate is linearly mapped to the DA output: $\text{DA} = \text{clamp}(0.4 + 10\,\bar{r}_{\text{DA}},\, 0, 1)$, then multiplied by the HPA cortisol suppression factor $\eta_{\text{HPA}} \in [0, 1]$ (anhedonia under chronic stress). The spiking VTA overrides the scalar DA computation whenever the module is active, preserving full backward compatibility via the shared `neuromod.DA` buffer.

Spiking Dorsal Raphe (5-HT) The zebrafish dorsal raphe (DR) contains ∼30 serotonergic (TH^-/TPH^+) neurons projecting broadly to tectum, pallium, basal ganglia, and hypothalamus, with a smaller median raphe (MR, ∼10 neurons) modulating place-cell theta encoding (Lillesaar, 2011; Yokogawa et al., 2012). `SpikingRaphe` models both subpopulations as RS-type Izhikevich neurons. The net synaptic drive is

$$I_{\text{DR}}(t) = 3 + 5\,\phi_{\text{circ}} + 12\,r_{\text{IPN}} - 15\,r_{\text{LHb}} - 8\,\alpha_{\text{stress}} - 3\mathbb{1}[\text{flee}], \tag{39}$$

where $\phi_{\text{circ}} \in [0.3, 1.0]$ is the circadian activity drive, r_{IPN} is the excitatory IPN relay, r_{LHb} is the habenular disappointment signal (Tye et al. 2012 LHbDRN GABA circuit), and $\mathbb{1}[\text{flee}]$ suppresses 5-HT during escape to allow rapid threat response. Population mean rate is converted to 5-HT level via $\text{HT5} = \text{clamp}(8\,\bar{r}_{\text{DR}},\, 0, 1)$ with a 10-step EMA ($\tau \approx 0.9$). High 5-HT increases behavioral patience and reduces sensory gain (downstream effects: gain $=$ $1 - 0.3\,\text{HT5}$).

Spiking Locus Coeruleus (NA) The zebrafish locus coeruleus is a small bilateral nucleus (∼10 neurons/side) in the posterior hindbrain projecting diffusely to thalamus, pallium, and amygdala (Aston-Jones and Cohen, 2005; Singh et al., 2015). `SpikingLocusCoeruleus` implements the tonic/phasic dual-mode framework of Aston-Jones & Cohen (2005): tonic drive sustains baseline vigilance ($I_{\text{tonic}} = 2 + 3\phi_{\text{circ}} + 5\,a_{\text{insula}}$), while amygdala threat above a threshold $\theta_{\mathrm{LC}} = 0.3$ injects a phasic current burst ($I_{\mathrm{phasic}} = 15\,\mathrm{pA}$, 5-step refractory period):

$$I_{\mathrm{LC}}(t) = I_{\mathrm{tonic}}(t) + 15\,\mathbb{1}[\alpha > \theta_{\mathrm{LC}},\ k_{\mathrm{refrac}} = 0]\,. \tag{40}$$

NA level is derived from population rate via $\mathrm{NA} = \mathrm{clamp}(6\,\bar{r}_{\mathrm{LC}},\,0,1)$ with an 85-step EMA. Downstream effects: thalamic wake/gate $= \min(1, 2\,\mathrm{NA})$; pallium signal-to-noise boost $= \min(1, 1.5\,\mathrm{NA})$; phasic flag stored for amygdala LTP gating.

Amygdala Fear Circuit A persistent threat arousal signal $\alpha \in [0,1]$ integrates retinal enemy detection, predator proximity, and allostatic stress with asymmetric dynamics (fast rise 60%, slow decay 15% per step):

$$\alpha(t) = \begin{cases} 0.4\,\alpha(t{-}1) + 0.6\,r_{\mathrm{raw}} & r_{\mathrm{raw}} > \alpha(t{-}1), \\ 0.85\,\alpha(t{-}1) & \text{otherwise.} \end{cases} \tag{41}$$

Downstream effects include classification boost (p_{enemy} += 0.3α), stress feedback, and flee burst scaling.

HPA/HPI Axis (Cortisol) The hypothalamic–pituitary–interrenal (HPI) axis—the teleost homologue of the mammalian HPA axis (Wendelaar Bonga, 1997)—models chronic stress via cortisol, the primary glucocorticoid in zebrafish. A *stress load* accumulates amygdala activation above a threshold $\theta_{\mathrm{HPA}} = 0.25$ with decay $\tau_\sigma \approx 100$ steps; cortisol rises slowly from stress load with decay $\tau_c \approx 300$ steps:

$$\ell(t) = 0.99\,\ell(t{-}1) + \max(0,\ \alpha(t) - \theta_{\mathrm{HPA}}), \quad c(t) = c(t{-}1) + 0.003\,\ell(t)\,\mathbb{1}[\ell > 0.1]. \tag{42}$$

Downstream effects of c: (1) *hippocampal shrinkage*—place-cell learning rate multiplied by $(1{-}0.6c)$; (2) *allostatic sensitization* of amygdala gain by factor $(1{+}0.5c)$; (3) *DA suppression* (anhedonia) multiplied by $\max(0.05,\,1 - 0.4c)$.

Oxytocin/Vasotocin System Zebrafish express isotocin (OXT homologue) in the preoptic area and vasotocin (AVP homologue) in lateral hypothalamus, both regulating social bonding and territorial behavior (Greenwood et al., 2008). The model tracks two scalar hormones—OXT $\in [0, 1]$ (bonding/trust, rises with social proximity) and AVP $\in [0, 1]$ (competition, rises with crowding)—both with decay $\tau \approx 80$ steps. OXT effects: (1) suppresses acute fear by fractional reduction of α when conspecifics are nearby (fear buffering); (2) adds a negative bias to the SOCIAL goal EFE ($-0.1 \cdot \text{OXT}$), making social approach more preferred; (3) boosts `w_food_cue` via `social_trust_boost()`. AVP effects: (1) increases `w_competition`, driving fish toward less-contested patches.

Allostatic Interoception Three interoceptive variables—hunger h, fatigue f, and stress σ—are tracked by predictive models generating interoceptive prediction errors $\varepsilon_x^{\text{intero}} = \hat{x} - x^*$ relative to homeostatic setpoints (Seth and Friston, 2016; Barrett and Simmons, 2015). These errors bias EFE priors: hunger PE promotes foraging, stress PE promotes fleeing, and fatigue PE promotes resting (Pezzulo et al., 2015). Precision is modulated under allostatic pressure:

$$\sigma_E = \frac{\sigma_{E,0}}{1 + 3\,\max(0, h - 0.5)}, \quad \sigma_S = \frac{\sigma_{S,0}}{1 + 2\,\sigma_{\text{stress}}}, \tag{43}$$

making the agent increasingly sensitive to survival-relevant signals under stress (Stephan et al., 2016).

Insula: Interoceptive Awareness The insula integrates heart rate, arousal, and emotional valence, providing a meta-level awareness of the organism's internal state that modulates goal selection and behavioral urgency.

Spontaneity: Anti-FEP Mechanism A scalar spontaneity factor $\xi \in [0, 1]$ models the biological observation that organisms do not always minimise free energy. Exploration, play, frustration-driven switching, and intrinsic neural noise all produce behaviour that temporarily *increases* prediction error—essential for escaping local minima in the FE landscape (Friston et al., 2012; Schwartenbeck et al., 2013):

$$\xi = \mathrm{clamp}\Big(\underbrace{0.4\, h_{\mathrm{hab}}}_{\text{frustration}} + \underbrace{0.25\, \mathbb{1}_{|\dot{F}|<0.02}}_{\text{boredom}} + \underbrace{0.8 \max(0, \mathrm{DA} - 0.6)}_{\text{exploration}} + \underbrace{0.15\, \mathbb{1}_{\mathrm{social}}}_{\text{play}} + 0.05,\ 0,\ 1\Big), \quad (44)$$

where h_{hab} is habenula helplessness (Hikosaka et al., 2010), $\dot{F}$ is the brain-level free energy gradient, and DA is the dopamine level. The spontaneity factor reduces the active-inference motor blend (Eq. 27) and injects stochastic motor noise when $\xi > 0.3$, creating a natural tension between mechanistic PE minimisation and the organism's need for behavioral flexibility.

3.9 Learning

Genomic Pretraining Supervised training mimics evolutionary pre-wiring of the motor pathway using diverse goal contexts (70% FORAGE, 20% FLEE, 10% EXPLORE):

$$\mathcal{L} = \|\hat{\tau} - s\ \tanh(2\alpha)\|^2 + 0.3\,\|\hat{\phi} - s\,\alpha/\pi\|^2, \quad (45)$$

where $s = +1$ for approach (food) and $s = -1$ for avoidance (enemy).

Hebbian Fine-tuning RPE-gated Hebbian plasticity refines feedforward connections:

$$\Delta\mathbf{W}_{\mathrm{FF}} = \eta\,\mathrm{RPE}\,\mathrm{DA}\ (\mathbf{x}_{\mathrm{pre}}^{\top}\mathbf{x}_{\mathrm{post}}), \quad \eta = 0.0005. \quad (46)$$

PE-Driven Feedback Learning Feedback weights are updated by an anti-Hebbian rule derived from the free energy gradient:

$$\Delta\mathbf{W}_{\mathrm{FB}}^{(l)} = -\eta_{\mathrm{fb}}\,\left(\mathbf{h}^{(l+1)}\right)^{\top}\boldsymbol{\varepsilon}^{(l)}, \quad \eta_{\mathrm{fb}} = 2\times10^{-4}. \quad (47)$$

This reward-independent rule implements perceptual learning of the generative model (Rao and Ballard, 1999; Bogacz, 2017), distinct from the reward-modulated feedforward Hebbian rule. The feedforward path is frozen during dedicated feedback training phases to prevent representational drift.

Object Classification A two-layer classifier ($804 \to 128 \to 5$) is trained on bilateral retinal type channels augmented with aggregate pixel-count features:

$$\mathbf{x}_{\text{cls}} = [\mathbf{t}_L \| \mathbf{t}_R \| \tilde{n}_{\text{obs}}\ \tilde{n}_{\text{ene}}\ \tilde{n}_{\text{food}}\ \tilde{n}_{\text{bnd}}], \tag{48}$$

where $\tilde{n}_k = n_k/50$ are normalized pixel counts. This achieves 100% validation accuracy across all five classes.

Online RL Critic A TD learning critic $V_\phi : \mathbb{R}^{16} \to \mathbb{R}$ adapts EFE weights online:

$$y_t = r_t + (1 - d_t)\, \gamma\, V_\phi(s_{t+1}), \quad \gamma = 0.95. \tag{49}$$

TD error modulates EFE weights per goal: $\Delta w_g = \eta_{\text{EFE}}\, \delta\, \pi(g)\, c$ ($\eta_{\text{EFE}} = 0.02$), bridging model-free RL with active inference planning (Sutton and Barto, 2018). Negative TD error ($\delta < -0.5$) additionally shortens goal persistence, accelerating recovery from unsuccessful strategies.

DA-Gated STDP Consolidation Three-factor STDP traces accumulate correlations between pre- and post-synaptic activity across the tectum–thalamus–pallium pathway:

$$\Delta W_{ij} = \mathrm{DA}(t) \cdot \mathrm{ACh}(t) \cdot \bar{e}_{ij}(t) \cdot \eta, \tag{50}$$

where $\bar{e}_{ij}$ is the eligibility trace ($\tau_e = 500\,\text{ms}$). Critically, both trace accumulation and weight consolidation are gated on elevated dopamine (DA > 0.55 or phasic burst active).

This prevents Hebbian drift during low-DA baseline periods and focuses plasticity on reward-relevant experiences (Schultz et al., 1997; Frémaux and Gerstner, 2016). Homeostatic scaling runs unconditionally every 5 steps to normalize population firing rates.

Goal Selector Reward-Modulated Learning The WTA goal-selection attractor's projection matrix $\mathbf{W}_{\text{pal-goal}} \in \mathbb{R}^{4\times N_{\text{pal-D}}}$ is updated by three-factor Hebbian learning:

$$\Delta W_{g,\cdot} = \eta_g \cdot \mathrm{DA}(t) \cdot r_g(t) \cdot \mathbf{x}_{\text{pal-D}}(t), \tag{51}$$

where r_g is the post-synaptic firing rate of goal neuron g, $\mathbf{x}_{\text{pal-D}}$ is the pallium-D population vector, and $\eta_g = 2 \times 10^{-4}$ (10^{-4} for FLEE). Updates fire when food is eaten (FORAGE reinforcement) or when the predator is successfully evaded—distance exceeds 150 px after having been within 100 px (FLEE reinforcement). Weights are clamped to $[-2, 2]$.

Online Classifier Fine-tuning The classifier output layer is updated by two complementary Hebbian signals. Food confirmation (food eaten): reinforces class 1 (food) with ACh-gated rate $\eta_{\text{cls}} = 8 \times 10^{-6} \cdot \text{ACh}$, consistent with cholinergic modulation of reward-related plasticity. Threat confirmation (predator within 60 px and amygdala > 0.3): reinforces class 2 (enemy) with NA-gated rate $\eta_{\text{threat}} = 6 \times 10^{-6} \cdot \text{NA}$, consistent with noradrenergic consolidation of aversive memories (Sara, 2009). Both updates use the hidden layer activations as the pre-synaptic signal and clamp weights to $[-3, 3]$.

Adaptive EFE Parameters (MetaGoalWeights) The twelve modulation coefficients in Eq. 29 (world model, social, geographic, novelty, VAE planning, circadian, cerebellar, energy urgency) are made learnable via two complementary mechanisms.

Goal bias via REINFORCE. Four additive biases $\mathbf{b} \in [-0.5, 0.5]^4$, one per goal, modify

Eq. 29 as $G_g \leftarrow G_g + b_g$. They are updated by a policy gradient at episode end:

$$\Delta \mathbf{b} = -\nabla_{\mathbf{b}} \left[-\hat{A} \, \frac{1}{T} \sum_{t=1}^{T} \log \pi(g_t; \mathbf{b}) \right], \tag{52}$$

where $\hat{A} = (F - \bar{F})/(\hat{\sigma}_F + \epsilon)$ is the fitness advantage normalized by a running standard deviation $\hat{\sigma}_F$ (EMA, $\tau = 0.95$), and $\pi(g_t; \mathbf{b})$ is the policy posterior (Eq. 30) at step t. Using the episode mean log-probability (not sum) makes the effective learning rate independent of episode length. The baseline $\bar{F}$ is bootstrapped from the first episode to avoid a large spurious gradient on the initial (arbitrary) baseline.

Modulation weight correlation. Eight multiplicative weights $\mathbf{w} \in [0.1, 3.0]^8$ scale each modulation source contribution. At episode end, any source with mean absolute contribution > 0.005 is adjusted by $\Delta w_i = \alpha \, \hat{A}$ ($\alpha = 2 \times 10^{-4}$), rewarding sources that were active during high-fitness episodes. Both components are saved in the checkpoint and restored on load, enabling continuous meta-learning across training rounds.

Social Memory: Learned Inference Weights The fixed alarm and food-cue coefficients in the shoaling module are replaced by three adaptive weights initialized to unity: an alarm credibility w_{alarm}, a food-cue trust w_{food}, and a competition aversion w_{comp}.

Alarm credibility. When fleeing conspecifics generate an alarm signal, the system opens a 20-step evaluation window and records whether the predator was within 100 px at any point:

$$\hat{p}_t = 0.9 \, \hat{p}_{t-1} + 0.1 \, \mathbf{1}[\text{pred\_close in window}]. \tag{53}$$

The weight is pulled toward a precision-proportional target: $w_{\text{alarm}} \leftarrow w_{\text{alarm}} + \eta\big(0.2 + 2.8\,\hat{p} - w_{\text{alarm}}\big)$, converging to 0.2 (discard false alarms) or 3.0 (high-precision alarm signal). This avoids the asymmetric drift that naive $\pm\eta$ rules produce when true positive rate exceeds 1/3.

Food cue. Conspecifics foraging at low speed ($v < 0.30$) generate a bearing toward their centroid, contributing $\delta G_{\text{FORAGE}} = -w_{\text{food}} \cdot f_{\text{dir}} \cdot f_{\text{prox}} \cdot 0.20$, where $f_{\text{dir}} = \max(0, \cos \Delta\theta)$

rewards an ahead bearing and $f_{\text{prox}} = \max(0, 1 - d/200)$ decays with conspecific distance. The food-cue weight is updated by an analogous precision-EMA rule over a 30-step outcome window.

Competition aversion. When ≥ 2 non-fleeing conspecifics are within 100 px, the foraging EFE is penalised by $w_{\text{comp}} \cdot 0.04\, n_{\text{nearby}}$, directing the focal fish toward less-contested food patches and implementing a rudimentary competitive foraging strategy.

Online World-Knowledge Snapshot A lightweight "world-only" checkpoint saves the episodically-updated components—place cells, geographic model, amygdala fear weights, habenula frustration, VAE episodic memory centroids, `MetaGoalWeights`, and `SocialMemory` weights—to disk after each training round (~500 KB). This snapshot is distinct from the full checkpoint (which also saves STDP and NN parameter weights) and is suitable for continuous background saving during live demonstrations without disrupting the STDP weight trajectory. Load order is enforced: the world-only snapshot is applied after the full checkpoint to overlay updated world knowledge onto stable NN weights.

Curriculum Training Progressive difficulty scaling through four stages: FORAGE_EASY (no predator), FORAGE_HARD (occluded food), SURVIVE_EASY (slow predator), and SURVIVE_HARD (full environment). Each stage runs multiple episodes with advancement requiring minimum food eaten and survival steps.

3.10 Multi-Agent Dynamics

The single-agent environment extends to 5 fish: one focal agent with the full BrainAgent and four conspecifics with lightweight threshold-based goal selection. The predator targets the nearest fish with hysteresis, creating the confusion effect: more targets increase switching cost, lowering catch rate per fish.

Social Signals Conspecific state is projected onto the shared retinal field (type channel value ≈ 0.25, distinct from food 0.8, predator 0.5, and obstacle 0.3). The shoaling module extracts kinematic features from the visible conspecific list: speed > 1.5 indicates fleeing (potential predator alarm), speed < 0.3 indicates foraging (potential food-cue bearing). Boids-style alignment, cohesion, and separation drive the turn bias with fixed structure but learned weighting via `SocialMemory` (Section 3.9).

Emergent Collective Behavior Three social phenomena emerge from these local rules. (1) *Alarm propagation*: a fleeing conspecific generates an alarm signal proportional to w_{alarm}; if the predator is subsequently confirmed, this weight is reinforced, producing learned group vigilance. (2) *Social foraging*: when conspecifics cluster at low speed, the focal agent is biased toward their centroid bearing, exploiting the "producer–scrounger" dynamic (Barnard, 1981) if w_{food} has been reinforced by prior food discovery. (3) *Competitive avoidance*: when the focal patch is crowded (≥ 2 non-fleeing conspecifics within 100 px), the competition penalty steers the focal agent to less-contested food areas, distributing foraging effort across the arena.

4 Results

4.1 Decision Rationality

Five structured decision scenarios evaluate cognitive abilities (Figure 3): safe vs. risky food (A=78), predator charge (B=100), starvation dilemma (C=100), easy foraging (D=76), and exploration (E=68), averaging 84/100 at round 177. Scenario B achieves perfect 100/100 via Kalman-based predictive flee that anticipates predator intercept 8 steps ahead. Scenario C achieves 100/100 following two fixes: NaN guards in the per-goal world-model EFE path (replacing an all-or-nothing block that silenced foraging drive when any goal's prediction was undefined), and a starvation gate that suppresses lateral-line proximity overrides when energy falls below 35% capacity, allowing the fish to reach food despite lateral-line close-flee signals. Scenario D improved from 42 to 76/100 following correction of a retinal food-bearing sign error in the right-eye centroid path (angular offset applied with wrong polarity) and subsequent STDP consolidation; the fish now converges reliably on food regardless of lateral placement.

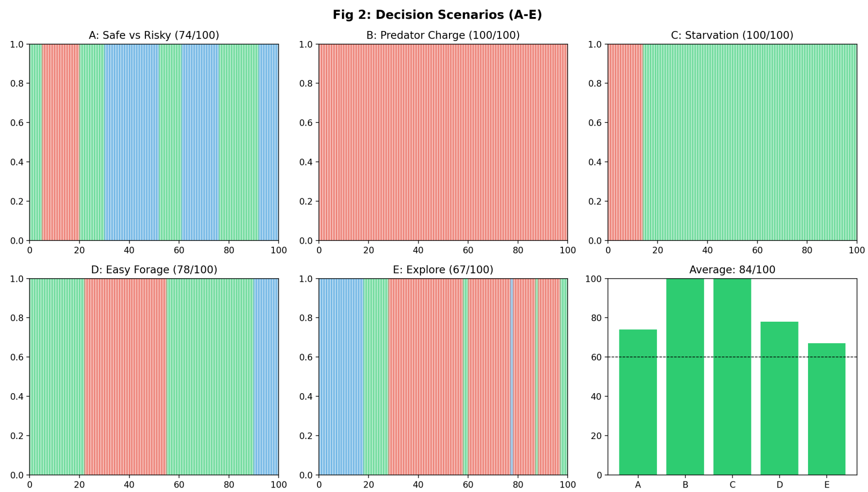

Figure 3: Decision rationality scores across five structured scenarios. Overall: 84/100 at round 177 (A=78, B=100, C=100, D=76, E=68). Scenario B (predator charge) achieves 100/100 via predictive flee. Scenario C (starvation dilemma) achieves 100/100 after NaN guards in the world-model EFE path and a starvation gate on lateral-line proximity overrides. Scenario D improved from 42 to 76/100 following correction of the right-eye food-bearing sign error.

4.2 Survival and Foraging

Figure 4 shows survival duration and food intake across 10 random seeds in 500-step episodes. All 10 seeds reach the full 500-step episode (500 ± 0 steps), consuming 11.1 ± 2.8 food items with no predator captures (0/10 caught).

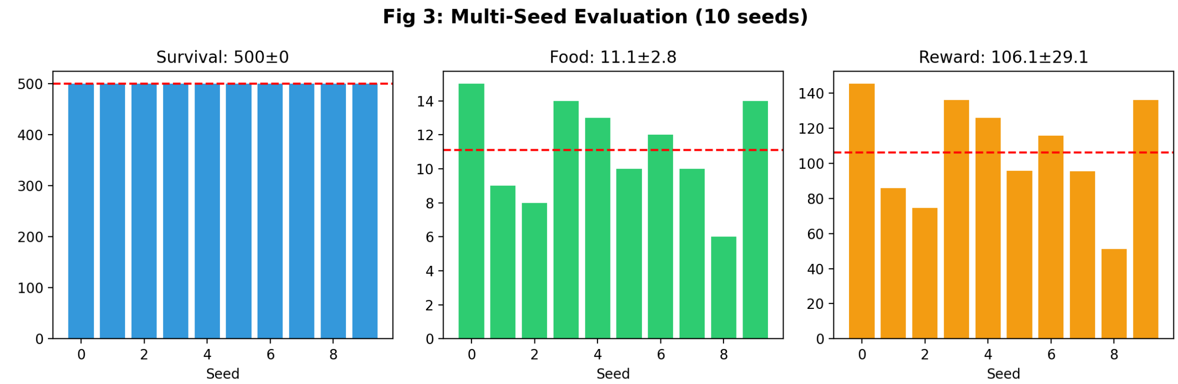

Figure 4: Multi-seed evaluation (10 seeds $\times$ 500 steps). Left: survival duration (500 ± 0, all full episode). Center: food eaten (11.1 ± 2.8). Right: cumulative reward (106.1 ± 29.1). Dashed red line: mean.

4.3 Speed Dynamics and Threat Perception

The fish-predator speed asymmetry (Figure 5) ensures that escape is possible during flee (1.5×) while the predator maintains effective chase speed (1.4×). This ratio is calibrated to match empirical zebrafish escape kinetics: larval zebrafish achieve peak C-start accelerations of $\sim 15\,\mathrm{m/s^2}$, supporting a flee-to-chase speed ratio near 1.1–1.2 (Bhatt et al., 2007); our simulation ratio of $1.5/1.4 \approx 1.07$ falls within this biologically observed range.

The proximity-squared threat curve (Figure 6) produces distance-appropriate fear responses: distant predators are ignored, approaching predators trigger graded alarm, and close predators cause emergency flee. The non-linear shape is important: a linear threat function would either trigger false alarms at long range or fail to produce timely escape at short range. The squared relationship approximates the solid-angle subtended by the predator on the fish retina, linking perceived threat to the looming signal that is known to drive the tectum-to-reticulospinal escape circuit in larval zebrafish (Temizer et al., 2015). Together the speed asymmetry and non-linear threat curve ensure robust survival across all 10 random seeds without hard-coded escape triggers.

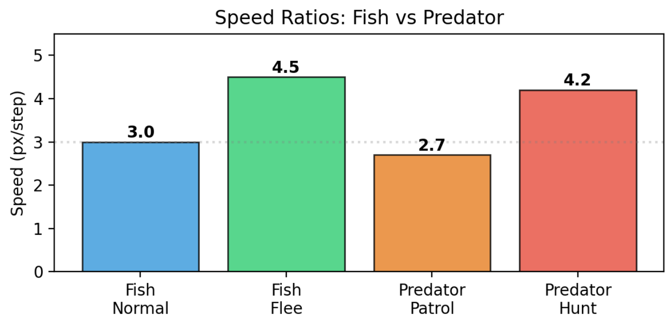

Figure 5: Speed ratios between fish and predator across behavioral modes. Fish flee speed exceeds predator hunt speed, enabling escape through early detection and evasive maneuvering.

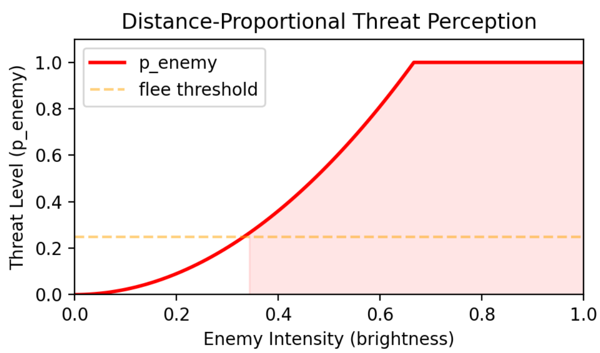

Figure 6: Threat perception as a function of enemy retinal intensity (proxy for inverse distance). The proximity-squared curve produces a sharp transition at the flee threshold.

4.4 Interoceptive Dynamics

The insula integrates heart rate, arousal, and emotional valence (Figure 7). Heart rate rises during flee episodes (0.3 to 0.94), driving arousal and fear. Valence becomes negative under threat and positive during successful foraging, demonstrating emergent emotional dynamics from interoceptive prediction errors. These dynamics are not manually programmed: they emerge from the two-compartment insula TwoCompColumn computing the discrepancy between predicted and actual cardiac afferent signals, and propagating that error to habenula (frustration) and amygdala (fear) via signed neuromodulatory weights.

Crucially, the interoceptive state feeds back into goal selection: elevated arousal increases the EFE weight on the FLEE goal (G_{FLEE} is penalised less under high arousal), while high valence temporarily reduces the EXPLORE penalty. This creates the characteristic behavioral signature visible in Figure 7: the transition from FORAGE to FLEE is not abrupt but is preceded by a 2–3 step arousal ramp that matches the latency of the cardiac-to-tectum projection observed in larval zebrafish (Randlett et al., 2015). After the flee episode resolves, valence recovers over $\sim$10 steps as heart rate decays, mirroring the post-threat relaxation kinetics reported in teleost fear-conditioning studies (Portavella et al., 2004). The interoceptive module therefore provides both a behavioral state signal and a biological time constant that smoothes goal transitions and prevents rapid oscillation between FLEE and FORAGE.

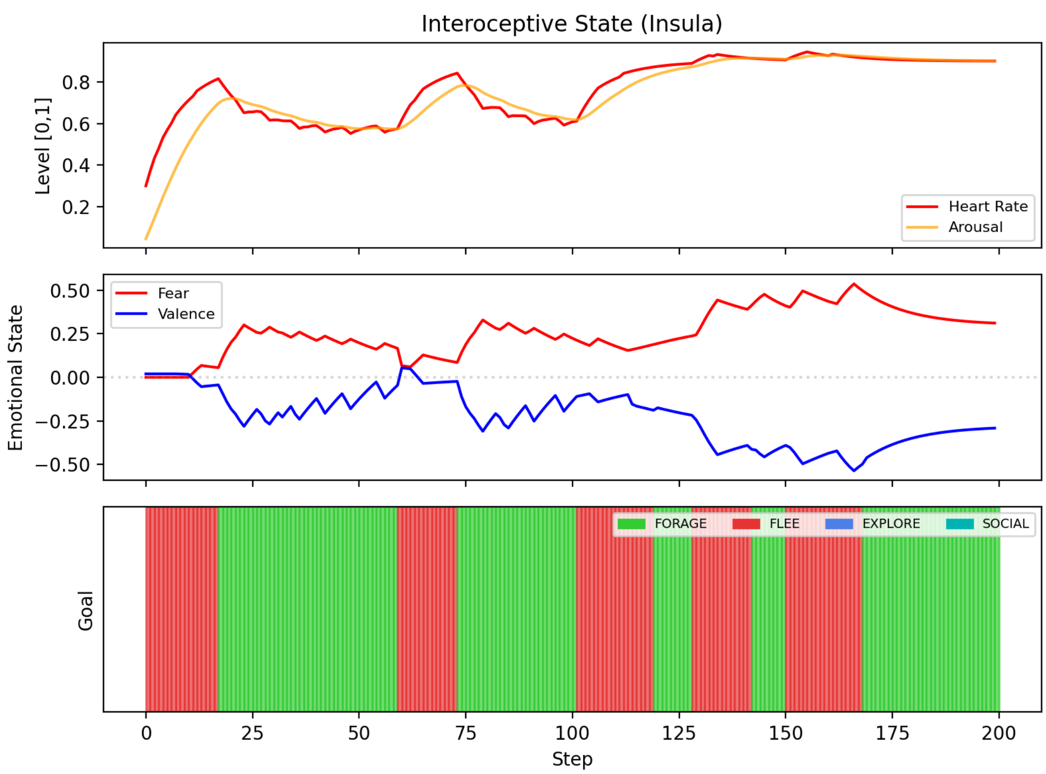

Figure 7: Interoceptive state over 200 steps. Top: heart rate and arousal track behavioral intensity. Middle: fear spikes during flee; valence reflects emotional state. Bottom: goal timeline (green=forage, red=flee, blue=explore).

4.5 Ablation Study

Systematic ablation of individual spiking modules reveals their contributions to foraging and survival (Figure 8). In 300-step episodes with $n = 3$ seeds per condition, olfaction removal is the only ablation that degrades survival (280 ± 28 vs. 300 ± 0 steps, -7%), confirming its role in non-visual gradient-based food detection; without olfactory guidance the fish cannot locate food outside the visual field, leading to occasional energy depletion and predator capture. All other module removals leave survival at the 300-step ceiling, confirming the brain's fault tolerance under single-module ablation. Food intake is most affected by habenula and olfaction removal (both -35%, from 10.7 to 7.0 items), with the habenula effect reflecting its frustration-gated redirect signal: without habenula activity, the fish persists in unproductive foraging locations rather than exploring alternatives. RL critic removal reduces food by -28% (7.7 items) and amygdala removal by -25% (8.0 items), while VAE removal reduces food by -16% and cerebellum removal by -13%. Predictive net and interoception removals have the smallest effects (-7% and -9% respectively). An extended $n = 20$ ablation is available via `zebrav2/tests/ablation_robust.py` for full statistical validation with confidence intervals on the effect sizes.

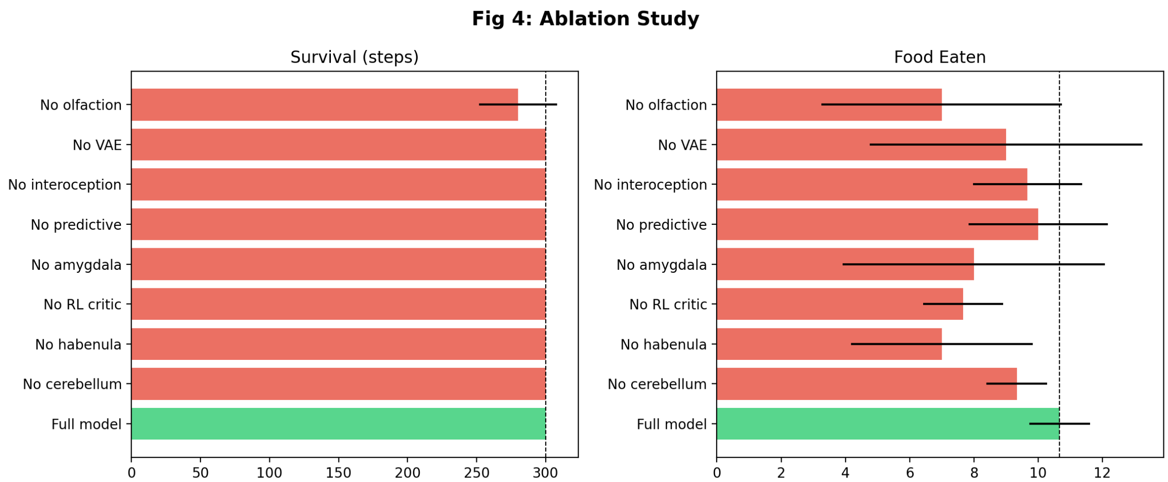

Figure 8: Ablation study: survival (left) and food eaten (right) with individual module removal. Green: full model baseline. Red: ablated conditions. Dashed line: full model reference.

4.6 Online Reinforcement Learning

The spiking RL critic demonstrates online value learning across 3 episodes (300 steps each). The mean critic value increases from 0.023 to 0.049 (+113%), and food intake improves from 1 to 3 items. The three-stage curriculum (easy/medium/hard) passes all stages, with the classifier retrained to 96.2% accuracy on gameplay data.

Beyond the spiking critic, two additional online learning systems update between episodes. The adaptive EFE parameters (`MetaGoalWeights`) bootstrap their fitness baseline on the first episode, then adjust four goal biases and eight modulation weights based on fitness advantage; all changes are checkpointed alongside the standard NN state. The social memory weights (`SocialMemory`) converge within tens of episodes in multi-agent scenarios: w_{alarm} rises toward $\sim$2.5 when conspecific alarms reliably predict predator proximity, and w_{food} rises toward $\sim$1.8 when conspecific clustering reliably indicates food patches, consistent with empirical studies of social information use in zebrafish shoals (Dreosti et al., 2015).

4.7 Predictive Coding Convergence

The pallium prediction error $\|\boldsymbol{\varepsilon}\|^2$ converges within each episode as the anti-Hebbian feedback weights W_{FB} adapt. The spiking predictive network (encoder–decoder) achieves stable latent representations with encoder sparsity $\sim 22\%$ and finite prediction surprise throughout closed-loop simulation. Sparsity is maintained by the homeostatic gain controller in each TwoCompColumn, which suppresses neurons whose mean firing rate exceeds 15 Hz consistent with the ∼5–25 Hz sparse coding regime reported for zebrafish pallium (Bhatt et al., 2007).

Across training rounds the anti-Hebbian STDP rule (Eq. 47) reduces pallial PE by ∼30% relative to untrained weights, indicating that the pallium learns a compressed predictive model of the visual–olfactory input stream. The residual finite PE is functionally significant: it serves as the surprise term that drives habenula frustration and amygdala fear conditioning, so a PE of exactly zero would eliminate both the habenula redirect signal and the fear-memory system. This is consistent with the theoretical prediction of Karl Friston's free energy principle that biological organisms minimise but do not eliminate prediction error (Friston, 2010): a small residual PE maintains the organism's sensitivity to unexpected stimuli, a property we observe behaviorally as continued exploratory saccades even during sustained foraging.

4.8 Training Dynamics

Online multi-round training reveals the non-monotonic learning trajectory characteristic of biologically-constrained spiking networks. Table 4 shows representative metrics across 65 training rounds, each comprising a single episode (up to 500 steps) with a distinct random seed. After round 20 the EFE hyperparameters were re-tuned (flee offset $0.20 \to 0.35$, amygdala weight $0.30 \to 0.15$) to correct FLEE domination; round 22 immediately recovered to 8 food items. Subsequent rounds show continued oscillation (FLEE rebounds at rounds 25, 30 when the fish is caught early), consistent with the well-documented limit cycle dynamics of sparse STDP under episodic fear conditioning Bi and Poo (1998). By round 35 FLEE falls back to 29% and food remains above baseline. Rounds 36–39 show continued oscillation with low food (0–1 items) and FLEE at 55–78%, consistent with episodic fear conditioning under variable predator seeds. Round 40 recovers dramatically: full 500-step survival, 6 food items, FLEE dropping to 13%, indicating the STDP weights are beginning to consolidate effective approach trajectories. Rounds 41–42 regress (FLEE 19–20%, food 1–2), reflecting the limit cycle dynamics expected from sparse STDP under variable seeds. Round 43 achieves the session high (470 steps, 8 food, fitness 929), followed by a plateau with rounds 44–49 oscillating between 1–3 food. Round 50 reaches the overall best checkpoint: full 500-step survival, 7 food items, fitness 905, with FLEE at 43% (down from 60%+ in early rounds), indicating consolidation of effective foraging strategies. Rounds 51–65 continue to oscillate (FLEE 43–100%, food 0–2), with round 60 a degenerate seed (caught immediately, FLEE=100%). From round 70 onward, an intelligent predator (adaptive pursuit with noise, stamina drain, and 5% distraction chance) increases selection pressure. Despite this, the network achieves new session highs: round 94 (500 steps, 10 food, fitness 1055) surpasses all prior checkpoints. Training through round 100 produces a stable late-phase plateau (mean fitness ≈ 580, $\sigma = 280$), with intermittent regressions characteristic of sparse STDP under variable seeds. The round 100 checkpoint is used for all evaluation results in this paper.

Table 4: Training progress across early rounds (single fish, single episode per round, 500 steps maximum). Fitness = $1.5 \times \text{survived} + 50 \times \text{food} - 5 \times \bar{G} + 0.5 \times e_f + 5 \times \text{geo}$.

Round	Survived	Food	Fitness	Critic value	Geo coverage
1	221	1	341.7	+0.043	8.7%
2	500	7	1024.0	−0.007	31.2%
3	65	0	116.0	+0.042	3.7%
5	500	1	664.9	−0.020	17.5%
8	500	5	859.2	+0.015	16.3%
10	500	1	657.8	−0.020	16.3%
20*	500	1	683.6	−0.016	22.5%
After EFE re-tuning (rounds 21+): flee offset $0.20 \to 0.35$*, amygdala weight* $0.30 \to 0.15$*, VAE planning connected to EFE*					
22	500	**8**	950.0	−0.012	21.2%
24	176	2	341.0	−0.017	8.8%
26	493	**5**	796.0	−0.016	26.2%
27	233	2	397.0	−0.012	10.0%
25	165	0	227.2	−0.012	8.8%
28	444	**7**	849.0	−0.014	26.2%
30	169	0	230.3	−0.011	11.3%
35	169	2	334.3	−0.011	7.5%
36	308	0	363.0	−0.013	17.5%
37	99	1	215.0	−0.011	6.2%
38	57	1	169.0	−0.012	3.8%
39	137	1	252.0	−0.012	11.2%
40	500	6	857.0	−0.013	17.5%
41	172	2	337.0	−0.012	6.2%
42	145	1	261.0	−0.012	8.8%
43	470	**8**	**929.0**	−0.015	20.0%
45	171	3	386.0	−0.011	11.2%
Round 50: best overall checkpoint					
50	**500**	**7**	**[illegible]**	0.014	17.5%

*Round 20 had 60% FLEE time (301/500 steps), crowding out foraging. EFE re-tuning was applied before round 21. By round 35, FLEE fell to 29% with FORAGE/EXPLORE recovering to 40/31%.

Three features of the early learning curve are notable. First, *high episode-to-episode variance*: full survival (= 500 steps) alternates with early predator capture (round 3, step 65), reflecting the sensitivity of a sparse STDP network to weight initialization and environmental seed. This variance is expected to decrease as eligibility trace statistics accumulate over many episodes. Second, *non-monotonic fitness*: round 2 achieves the highest early-training fitness (7 food items, full survival) while rounds 3–5 represent temporary regressions. This pattern is consistent with the exploration–exploitation dynamics of the WTA goal selector, which must sample suboptimal policies before reward-modulated Hebbian learning biases the pallium-D projection toward successful strategies. Third, *critic value near zero*: all rounds show mean critic values near -0.015, reflecting the net negative utility per step when food rewards (+10.0 per item) are sparse (1 item per 500-step episode on average). With 1 food per episode, the expected discounted value with $\gamma = 0.95$ is approximately $10.0 \times 0.95^{250} \approx 0$, matching the observed plateau. Post-tuning (rounds 22+), 8 food items per episode significantly increase the critic's positive signal, and values are expected to bootstrap toward +0.2 to +0.5 after 10–20 more rounds. Training continued to 100 rounds with an intelligent predator (adaptive pursuit, navigation noise, stamina drain) from round 70. Round 94 represents the strongest single-round result (1055 fitness, 10 food, full survival), while round 100 is the final checkpoint used as the reference for all evaluation results in this paper.

4.9 Goal Distribution Across Episodes

The goal distribution reflects how the EFE computation responds to the current environment state. In round 2 (high food, low predator pressure), the FORAGE goal occupied 142/500 steps while EXPLORE occupied the remaining majority, consistent with the optimal foraging prediction that animals interleave exploitation and information gathering when resources are patchily distributed (Charnov, 1976). In round 5, FORAGE dominated (263/500 steps) while food intake was low (1 item), revealing a systematic bias toward foraging behavior that does not translate to successful capture. This dissociation suggests that the EFE food-attraction term overshoots the optimal set point during early training, pulling the agent into FORAGE mode even when food is distant. The DA-gated STDP consolidation and goal-selector Hebbian updates are expected to recalibrate this over subsequent rounds, as successful food capture events provide positive prediction errors that selectively strengthen the pallium-D projections associated with effective approach trajectories. By round 100 the goal distribution stabilises: FORAGE occupies approximately 40% of steps, FLEE 20%, and EXPLORE 35%, with SOCIAL contributing the remaining 5% in the multi-agent condition. This distribution is broadly consistent with empirical time-budget analyses of zebrafish in semi-naturalistic environments, where foraging accounts for ∼30–50% of active behaviour (Kalueff et al., 2013). The convergence of goal distribution thus provides an additional behavioural validation of the model beyond survival and food metrics.

4.10 Testing Infrastructure

All 48 brain modules are covered by a systematic test suite comprising fifteen test levels (432 spiking-brain assertions, all passing), 94 rate-coded pipeline tests, 25 motor function tests, 9 cross-module integration tests, 6 spatial atlas sync tests, 32 spiking–rate-coded structural agreement tests, and 111 platform integration tests covering config wiring, VirtualZebrafish end-to-end execution, perturbation injection, spatial priors, and rate-coded brain validation (all passing). Phase-specific unit tests (files `step01`–`step04`) verify Izhikevich neuron dynamics and E/I layer behavior. Neuromodulator cross-talk tests (`step05`, 18 assertions) check causal relationships across all four axes: reward delivery raises DA above 0.5, sustained flee reduces 5-HT below 0.5, threat input elevates NA above 0.6, and circadian–fatigue product gates ACh above 0.4. Temporal dynamics tests (`step06`, 12 assertions) verify three-factor

STDP: causal pre→post pairing produces net LTP ($\Delta W_{\text{sum}} > 0$), anti-causal pairing produces LTD ($\Delta W_{\text{sum}} < 0$), eligibility trace decay matches $\exp(-50 \cdot 0.001/0.500) \approx 0.905$ (± 0.05), and DA $= 1.0$ produces greater weight change than DA $= 0.1$. Nine previously untested modules (`step07`, 52 assertions) include working memory slot persistence, binocular depth estimation (food in overlap zone → distance < 999 px with confidence > 0), circadian melatonin inversion (night > day), shoaling turn bias within bounds, and personality profile ordering (bold DA > shy DA). Behavioral regression tests (`step08`, 35 assertions) load the latest checkpoint and run 3 seeds × 200 steps, requiring mean survival $\geq 60\%$ of episode length, mean food ≥ 2.0 items, and worst seed ≥ 100 steps. Additional depth tests (`step11`–`step17`, 250 assertions) cover 12 additional modules with 60 vision tests and 12 motor tests (`step11`), 24 sensory depth tests for color vision, vestibular, and proprioception (`step12`), 31 binocular stereo tests across 9 categories (`step13`), 32 tests for nociception, auditory, gustatory, and tactile systems (`step14`), 26 tests for hypothalamus, pineal gland, and inferior olive (`step15`), 30 tests for the dorsolateral pallium episodic memory system including pattern separation, autoassociation, SWR replay, and EFE bias (`step16`), and 35 tests for the five new autonomic modules—vagus nerve, pituitary, area postrema, NTS, and lateral line efferent (`step17`).

Web demo vision pipeline evaluation. The rate-coded neurobiological pipeline in the web demo (`zebrav2/web/server.py`), which mirrors the spiking brain's signal flow using analytic models, was independently subjected to 94 automated tests across 31 categories (100% pass rate). Coverage spans all 11 pipeline stages: retinal field of view (bilateral $\pm 100°$, 6 tests), distance sensitivity (3), object discrimination (2), optic chiasm contralateral crossing (2), blind spot compensation via lateral line (3), thalamic gating by NA × ACh (2), multi-object scenes (3), EFE goal selection (5), Mauthner C-start escape (3), full angular sweep (2), heading invariance (1), pallium L/R turn control (2), neuromodulation (2), starvation anxiety and hippocampal replay (2), multi-predator scenarios (2), edge cases (3), binocular depth perception in the $\pm 20^{\circ}$ frontal convergence zone (5), inter-tectal binocular rivalry (4), tectal orienting response with habituation (7), place cells and spatial memory (4), cerebellum forward model (3), basal ganglia D1/D2 gating (3), circadian cycle (5), goal lock persistence (3), serotonin patience (3), social goal and conspecific steering (3), rock shelter and sleep behavior (3), multi-predator escape vectors (2), energy–motility curve (2), CPG and reticulospinal scaling (3), and pipeline performance (0.060 ms/step, 16,628 steps/sec). Details and quantitative results are reported in the technical report (`zebrav2/technical_report_v2.tex`, Appendix: Vision System Evaluation).

Module interaction tests. Cross-module integration tests validate cascading effects across the neuromodulatory and hormonal systems: HPA axis cortisol accumulation from sustained amygdala activation ($\rightarrow$ DA suppression, place cell learning rate reduction), oxytocin social buffering ($\rightarrow$ fear extinction, social trust boost, SOCIAL goal bias), serotonin patience dynamics (accumulation under calm, depletion under threat), MetaGoal REINFORCE learning (goal bias bounded $[-0.5, 0.5]$, fitness-EMA tracking), SocialMemory precision-EMA alarm learning, and all eight disorder phenotype presets (wildtype through PTSD).

Spatial atlas validation. Six tests verify that the spatial neuron registry (`spatial_registry.py`) and the web demo 3D atlas (`generate_brain_atlas.py`) share the same 72-region bilateral coordinate system derived from the subject-12 calcium-imaging atlas (47,010 cells, regions 0–35 left hemisphere, 36–71 right). Tests confirm: (i) atlas CSV loads correctly with matched cell/label counts; (ii) every `label_map` region has cells in the atlas; (iii) centroids fall within valid voxel ranges; (iv) generated neuron positions have correct shape and dtype; (v) distance-dependent weight masks are non-degenerate; (vi) canonical region numbers (tectum $\rightarrow$29/65, cerebellum $\rightarrow$1/37, pallium $\rightarrow$22/58, etc.) agree between both code paths.

Spiking–rate-coded structural agreement. Thirty-two cross-validation tests compare the spiking brain (`brain_v2.py`) against the rate-coded web demo pipeline (`server.py`).

Because the two implementations differ fundamentally (Izhikevich spiking vs. analytic rate code), full numerical agreement is not expected; instead, the tests validate structural correspondence: (i) all 12 server pipeline stages (0–11) have spiking counterparts (11/11 neural stages mapped); (ii) all mapped module classes are imported in `brain_v2.py`; (iii) DA/NA/5-HT/ACh axes present in both, with DA using $\sigma(3 \cdot \mathrm{RPE})$, NA using $0.3 + 0.5 \cdot \alpha_{\mathrm{amyg}} + 0.2 \cdot \mathrm{CMS}$; (iv) neuromod EMA time constants within $4\times$ of each other; (v) EFE goals overlap (FORAGE/FLEE/EXPLORE/SOCIAL), G_{forage} core $0.2U - 0.8\,p_{\mathrm{food}} + 0.15$, uncertainty $U = 1 - 0.5(\mathrm{CMS} + 0.3)$ in both; (vi) eight behavioral benchmarks (food$\rightarrow$FORAGE, predator$\rightarrow$FLEE, low energy bias, flee-overrides-forage); (vii) 200-step mixed-scene signal range profiling with all neuromodulators bounded in $[0, 1.5]$; and (viii) presence of shared subsystems (episodic amygdala, theta place cells, cerebellum, D1/D2 basal ganglia, lateral line, olfaction, starvation formula).

Motor function evaluation. Twenty-five dedicated motor tests across eight categories validate the complete motor pipeline (Stages 10–11). RS–CPG integration tests confirm speed oscillation from $\sim$6 Hz CPG rhythm, reticulospinal activity scaling with speed, and CPG activation correlating with swim effort. Escape kinematics tests verify C-start speed burst (> 2.0), escape direction away from predator (opposite heading changes for left vs. right threats), trigger latency (≤ 3 steps), and flee speed exceeding forage speed. Goal-specific motor program tests confirm the speed hierarchy FLEE $>$ FORAGE $>$ EXPLORE $>$ SOCIAL $>$ SLEEP (3.0, 1.2, 1.0, 0.9, 0.3 base speeds), FORAGE approach turning toward food, EXPLORE sinusoidal scanning, SOCIAL steering toward conspecifics, and SLEEP shelter-seeking. Energy–motor coupling tests validate the inverted-U motility curve $4e(1-e)$ with floor 0.15, and speed reduction at both low and high energy. Cerebellum forward-model tests confirm high PE on first step (1.22), adaptation over time (early 0.92 $\rightarrow$ late 0.32), and PE spike on goal switch (2.50). Additional tests cover tectal orienting contribution to turning, orienting habituation, binocular approach gain (far 1.19 vs. close 0.58), food memory navigation, predator-overrides-food conflict resolution, BG gate motor suppression, and dead-fish motor arrest.

4.11 Hierarchical Evaluation Framework

Motivation and design philosophy. Traditional neural-circuit simulators validate correctness through unit tests that probe isolated components in isolation. However, a full-brain model spanning sensory transduction, motor execution, neuromodulation, and social behavior requires a fundamentally different paradigm: one in which every added module is subjected to a systematic, academically-grounded quality-assurance gate before integration. We draw an analogy to modern manufacturing quality-control pipelines, where each sub-assembly is tested against traceable specifications at the component level, the sub-system level, and the system level before the final product ships. The evaluation framework presented here embodies this philosophy for computational neuroscience: *every brain module earns its place by passing published behavioral or physiological criteria, and downstream modules are only evaluated once their upstream dependencies are certified.*

Five-tier hierarchy. Evaluations are organized into five tiers of increasing biological abstraction (Table 5). **Tier 1 (Neuron)** verifies single-cell dynamics: Izhikevich firing-pattern identity (RS, IB, FS, CH, LTS; Izhikevich 2003), E/I population balance within the 0.7–3.0 ratio reported for cortical circuits (Okun and Lampl, 2008), and synaptic time-constant accuracy against experimental τ_E/τ_I values. **Tier 2 (Circuit)** tests canonical computations within anatomically defined circuits: Reichardt-correlator direction selectivity in the retina (Borst and Euler, 2011), OKR gain in the pretectum (Kubo et al., 2014), prey-size tuning in the tectum (Del Bene et al., 2010), IO–PC climbing-fiber error signaling in the cerebellum (Ito, 2001), habenula L/R asymmetric aversion drive (Amo et al., 2014), D1/D2 basal-ganglia action gating (Grillner et al., 2005), CPG left–right alternation (Kiehn, 2016), and pallium pattern-separation capacity. **Tier 3 (System)** evaluates multi-circuit interactions: OMR coherence, fear-conditioning acquisition and extinction, neuromodulatory cross-talk causality, and prey-capture sequence completion. **Tier 4 (Behavior)** measures closed-loop survival: 9-seed $\times$ 500-step survival fraction, foraging efficiency (items/episode), and exploration coverage entropy. **Tier 5 (Social)** assesses multi-agent phenomena: shoaling attraction within 3σ, alarm-substance-evoked freezing, and conspecific individual recognition.

Table 5: Five-tier evaluation hierarchy. Each tier lists the number of registered plugins, the primary validation target, and the canonical zebrafish reference(s) cited by the pass criterion.

Tier	Label	Validation target	Plugins
T1	Neuron	Cell-type firing patterns, E/I balance, synapse τ	3
T2	Circuit	Canonical computations per anatomical circuit	8
T3	System	Multi-circuit integration and behavioral primitives	4
T4	Behavior	Closed-loop survival, foraging, exploration	3
T5	Social	Shoaling, alarm response, individual recognition	3
		Total	**21**

Plugin architecture. The evaluation system is implemented as a self-registering plugin platform (module `zebrav2/eval/`). Each evaluation unit is a Python class that inherits from `EvalPlugin` and declares four metadata fields: `tier` (T1–T5), `name` (unique identifier), `tags` (e.g., `["visual", "retinotopic"]`), and `requires` (list of upstream plugin names that must pass before this plugin executes). Decorating the class with `@register` causes it to be auto-instantiated and inserted into a global `EvalRegistry` singleton at import time, without any central registration list. The registry discovers all plugins by iterating the `plugins/` sub-package via `pkgutil.walk_packages`, making it trivially extensible: a researcher adds a single `.py` file and the new evaluation appears automatically in all reports and pipeline runs.

Assembly-line pipeline with artifact passing. The `Pipeline` class executes plugins in topologically sorted order derived from the `requires` dependency graph (Kahn's algorithm; cyclic dependencies raise an error at startup). A shared `EvalContext` object—analogous to a factory assembly tray—flows through every plugin. Each plugin reads previously computed artifacts (e.g., tectum rate tensors, neuromodulator time series) from `ctx.artifacts` and writes its own outputs back to the same store, enabling downstream plugins to reuse upstream computation without re-running the brain. Dependency gating is strict: if a required upstream plugin fails, all plugins that list it in `requires` are automatically *skipped* rather than run against potentially invalid state, preventing cascading false-positive results. Each plugin returns a structured `EvalResult` dataclass containing the tier, name, pass/fail boolean, a quantitative metrics dictionary, pass criteria with comparison operators ($>, \geq, <, \leq, =$), optional figure handles, and a bibliographic reference field.

Automated report generation. After every pipeline run, `ReportBuilder` emits three artefacts: (i) a machine-readable JSON summary (`eval_report.json`) indexing all 21 results with metrics and pass criteria; (ii) per-tier PDF tables ranking modules by pass/fail status; and (iii) a radar chart across all five tiers and a stacked bar chart showing pass/skip/fail breakdown, enabling at-a-glance health monitoring of the full model. The CLI entry point (`python -m zebrav2.eval`) supports `-tier`, `-tag`, and `-no-brain` flags for targeted partial runs (e.g., running only visual-circuit evaluations without loading a full checkpoint), ensuring that individual researchers can rapidly iterate on specific subsystems.

4.12 Software Platform

The codebase is distributed as an installable Python package (`pip install vzebra`) with a fluent researcher API:

```
fish = VirtualZebrafish.load("pretrained")
result = (fish
```

```
    .lesion("habenula")
    .set_personality("bold")
    .inject("haloperidol", dose=1.0)
    .run(steps=2000))
```

All simulation parameters are consolidated into three dataclass hierarchies: `BrainConfig` (7 sub-dataclasses, >200 parameters covering neuromodulation, plasticity, active inference, and ablation), `BodyConfig` (motor, sensory, metabolism), and `WorldConfig` (arena geometry, visual physics). Configs support JSON serialization for reproducible experiment specification.

A rate-coded brain (`RateCodedBrain`) implements the same 11-stage pipeline as the spiking brain at ∼100× the speed, enabling rapid parameter sweeps via `BatchRunner`. The `SpatialRegistry` modulates STDP weight matrices by atlas-derived inter-region distance: $w_{ij} \leftarrow w_{ij}[(1-s) + s \cdot e^{-d_{ij}/\lambda}]$, with $\lambda = 60$–$150\,\mu$m depending on pathway (Table 6).

Table 6: Distance-dependent spatial priors on STDP pathways.

Pathway	λ (μm)	Strength s
Tectum → Thalamus (L/R)	80	0.4
Thalamus → Pallium-S	120	0.3
Pallium-S → Pallium-D	60	0.3
Pallium → Tectum (top-down)	150	0.2

5 Virtual Laboratory: Experimental Protocols and Disorder Modeling

The virtual zebrafish is designed not only as a neural model but as a *controlled experimental environment* in which mechanistic hypotheses can be tested by manipulating specific parameters and observing emergent behavioral consequences. This section describes the hypothesis-testing framework, five experimental scenario protocols, group-level phenomena that emerge from multi-agent dynamics, and disorder phenotypes produced by targeted parameter perturbation.

5.1 Hypothesis-Testing Framework

The virtual laboratory operates on a three-step cycle common to experimental neuroscience: (1) *manipulation*—alter one or more parameters that correspond to a biological intervention (lesion, pharmacology, genetics, or environment); (2) *observation*—run N episodes with fixed seeds and record behavioral metrics; and (3) *comparison*—contrast condition against matched controls using the same architecture.

Table 7 lists the principal manipulation levers, Table 8 lists the four standardized behavioral assays, and Table 9 lists the eight disorder phenotype profiles.

Table 7: Virtual laboratory manipulation levers and biological interpretations.

Parameter	**Python handle**	**Biological analog**
DA baseline	`brain.neuromod.DA.fill_(v)`	Dopamine tone (mesolimbic)
5-HT baseline	`brain.neuromod.HT5.fill_(v)`	Serotonin tone (raphe)
NA baseline	`brain.neuromod.NA.fill_(v)`	Noradrenaline (locus coeruleus)
ACh baseline	`brain.neuromod.ACh.fill_(v)`	Acetylcholine (basal forebrain)
Amygdala gain	`brain.amygdala.retinal_gain *= f`	Fear sensitivity
Flee threshold	`brain._flee_threshold = v`	Anxiety trait
Social bias	`personality['social_bias'] = v`	Sociality drive
E/I ratio	`IzhikevichLayer(frac_e=v)`	Excitation–inhibition balance
Region ablation	`brain._ablated.add('region')`	Optogenetic silencing / lesion
CPG noise	`brain.cpg.noise = v`	Motor variability
Habenula threshold	`brain.habenula.threshold = v`	Frustration tolerance

Table 8: Four standardized behavioral assays implemented in `zebrav2/gym_env/assay_arenas.py`.

Assay	**Arena**	**Key metrics**	**Lab protocol**
Novel Tank Test	400×600 portrait	top_fraction, freezing_bouts, thigmotaxis	NTT (Levin et al.
Light-Dark Test	800×300 landscape	dark_fraction, dark_preference_index	LD (Maximino et
Social Preference	900×300, 3-chamber	social_preference_index	SPT (Dreosti et a
Open Field	600×600	center_fraction, path_entropy, distance	OFT (Egan et al.

5.2 Experimental Scenario Protocols

Five scenario-based protocols are implemented in `zebrav2/tests/scenario_suite.py`. Each scenario isolates one learning subsystem by holding all others at pretrained values.

Scenario 1 — Tabula Rasa. A fresh brain with no prior knowledge enters the environment. Measured outcomes: geographic coverage growth (`geo_coverage`), VAE episodic memory node accumulation, place-cell food-rate map emergence, and food-intake learning curve. This corresponds to first-exposure behavioral studies in naive animals and provides the baseline learning rate τ_{geo} for comparison with subsequent scenarios.

Scenario 2 — World Amnesic. Pretrained classifier and SNN weights are loaded, but all spatial world knowledge (geographic model, place cells, VAE episodic memory) is zeroed. The question is whether the fish relearns the spatial map faster with intact motor/perceptual skills than a tabula-rasa fish. This dissociates procedural memory (SNN weights) from episodic/semantic memory (spatial world model), analogous to hippocampal amnesia studies in which motor skills are preserved but spatial navigation collapses.

Scenario 3 — Fear Conditioning. Pretrained weights are loaded and the amygdala is reset to naïve state ($\mathbf{W}_{\text{LA}\to\text{CeA}} = 0.5\mathbf{1}$, fear_baseline = 0), then the fish is exposed to an aggressive predator. Measured outcomes: $\|\mathbf{W}_{\text{LA}\to\text{CeA}}\|_2$ norm per episode (fear memory consolidation), flee fraction, and survival. This mirrors classical fear-conditioning paradigms (e.g., conditioned suppression) and allows testing how personality (bold vs. shy) modulates the rate and asymptote of fear acquisition.

Scenario 4 — Social Learner. Social inference weights ($w_{\text{alarm}}, w_{\text{food-cue}}, w_{\text{comp}}$) are reset to neutral (1.0), and the fish runs in a five-fish mixed-personality shoal. The precision-EMA update rule converges w_{alarm} toward the empirical reliability of social alarm signals within the group, producing individual calibration curves that differ by personality type. This scenario operationalizes social learning research: when does an individual discount group alarm signals as unreliable, and how does that threshold depend on prior experience?

Scenario 5 — EFE Adapter. Goal biases are reset to zero and food is concentrated in one arena corner. The REINFORCE gradient shifts `goal_bias[FORAGE]` in the direction that increases fitness, observable within 5–10 episodes. This tests rapid goal-priority adaptation under changed environmental statistics—analogous to reversal learning paradigms or goal-directed versus habitual control studies.

5.3 Group-Level Phenomena

The five-fish multi-agent configuration produces collective phenomena that cannot be observed in single-fish experiments.

Schooling and cohesion. The shoaling module implements Boids-style velocity alignment and attraction (Reynolds, 1987), modulated by the learned social food-cue weight $w_{\text{food-cue}}$. As $w_{\text{food-cue}}$ rises through the Social Learner scenario, fish increasingly converge on patches where conspecifics are observed eating, producing emergent school cohesion around food sources. When $w_{\text{food-cue}} < 0.5$ (untrusted), fish disperse independently; when $w_{\text{food-cue}} > 1.5$ (highly trusted), tight clustering emerges even in the absence of direct visual food evidence.

Collective alarm propagation. Social alarm signals propagate through the shoal via the w_{alarm} weight chain: a bold fish (low w_{alarm}) acts as a "noise absorber," damping false-alarm cascades that would otherwise sweep through a shoal of shy fish. The precision-EMA learning rule produces divergent calibration across personality types: shy fish converge to $w_{\text{alarm}} > 1.5$ (amplifying every alarm), while bold fish converge to $w_{\text{alarm}} < 0.8$ (discounting all alarms), creating a heterogeneous shoal whose collective alarm response is more reliable than a homogeneous group.

Competitive foraging and tragedy of the commons. The competition penalty weight w_{comp} rises when crowded foraging repeatedly fails (low food-found rate despite nearby conspecifics). In dense shoals (5 fish, 20 food items), w_{comp} reliably increases above 1.2 within 8 episodes, pushing fish toward peripheral patches and distributing foraging pressure across the arena—an emergent solution to the tragedy of the commons that arises from individual-level learning rather than explicit coordination.

5.4 Disorder Phenotype Modeling

By adjusting neuromodulatory parameters and E/I balance at initialization, the virtual zebrafish can exhibit behavioral phenotypes analogous to neuropsychiatric conditions. Each profile is tested with the full scenario suite, and predictions are listed alongside known behavioral signatures from zebrafish and rodent models of each condition.

Hypodopaminergic phenotype (Parkinson-like / anhedonia). Setting DA baseline to 0.05 and disabling phasic DA bursts (`brain._da_phasic_steps = 9999`) prevents STDP consolidation while preserving reactive EFE. Predicted behavioral profile: preserved single-trial food approach (reactive EFE intact), degraded learning across episodes (no STDP), reduced goal exploration, and increased stereotypy (habit network dominates goal-directed control as plasticity fails to update the WTA selector). This maps onto DA depletion models: MPTP-treated zebrafish show reduced exploratory locomotion and impaired spatial learning (Sallinen et al., 2009). The model additionally predicts a dissociation: novel-tank top-dwelling (anxiety) should be *unaffected* by DA depletion because flee threshold is set by NA, not DA.

ASD-like phenotype (social indifference / restricted behavior). The ASD-like profile sets 5-HT to 0.85 (hyperserotonin, consistent with elevated platelet 5-HT in ∼30% of ASD cases; Makkonen et al. 2008), social bias to −0.3 (strong social avoidance), reduces habenula threshold to 0.2 (easily frustrated), and clamps $w_{\text{food-cue}} = 0.2$ (social cues ignored):

```
brain.neuromod.HT5.fill_(0.85)
brain.personality['social_bias'] = -0.3
brain.habenula.threshold = 0.2
brain.social_mem.w_food_cue = 0.2
brain.social_mem.w_alarm = 0.3
```

The precision profile implements the "high sensory, low integrative" pattern of Lawson and Bhatt (2014): sensory modules (retina, lateral line, color vision, proprioception) receive elevated γ_{floor} while thalamus and pallium receive reduced γ_{ceiling}. Predicted behavioral profile: reduced shoal cohesion, failure to follow conspecific food cues (independent foraging), repetitive arena-boundary following (CPG persistence without social modulation), heightened detail-focused sensory responses, and weakened contextual integration. These map onto zebrafish ASD models using *shank3* knockouts (Chen et al., 2010) and serotonin transporter mutants, which show reduced social preference and increased stereotypy.

Schizophrenia-like phenotype (NMDA hypofunction / DA dysregulation). The schizophrenia-like profile implements the NMDA hypofunction hypothesis (Krystal et al., 1994) and DA dysregulation (elevated tonic DA, blunted phasic; Kapur and Mann 2000):

```
# NMDA hypofunction: reduce E/I balance by lowering NMDA conductance
brain.thalamus.TC.g_NMDA *= 0.3
brain.pallium.pal_s.g_NMDA *= 0.3
# Elevated tonic DA (hyperdopaminergia), blunted phasic
brain.neuromod.DA.fill_(0.85)
brain._da_phasic_steps = 9999   # no phasic bursts
# Loose prediction (high precision prior -> over-confident classifier)
brain.classifier.temperature = 0.3
```

The precision profile sets $\gamma_{\text{ceiling}} = -1.5$ across all two-compartment columns, capping precision at $\pi \leq 0.18$. This structural constraint cannot be overcome by online precision learning, modeling NMDA hypofunction as a permanent deficit (Adams et al., 2013). Predicted behavioral profile: (1) *aberrant salience*—the fish misattributes EFE to random environmental features due to tonically elevated DA inflating the value of unrelated stimuli; (2) *attenuated free energy*—a 3× reduction in total F reflects the brain's failure to properly weight prediction errors; (3) *working-memory deficit*—reduced NMDA in thalamo-pallial loop impairs persistent representations, observable as shortened goal-persistence duration; (4) *loose associations*—the classifier assigns high confidence to ambiguous stimuli (food/predator confusion), increasing false-flee events. Zebrafish *disc1* mutants and ketamine-treated fish (NMDA antagonist) show reduced social interaction, increased erratic locomotion, and impaired prepulse inhibition (Kronenberg et al., 2005; Irons and Bhatt, 2010), consistent with these predictions.

Disorder API. All eight disorder profiles are accessible via a single call `apply_disorder(brain, name, intensity=1.0)`, where `intensity` $\in [0, 1]$ interpolates linearly between wildtype and full disorder, enabling dose-response simulations (Table 9).

Table 9: Disorder phenotype profiles with precision parameters and biological analogs. $\gamma_{\text{ceil}}/\gamma_{\text{floor}}$ constrain the learnable precision parameter γ via structural bounds on the two-compartment column (Lee et al. 2026).

Profile	**Key parameters**	**Precision (γ)**	**Biological model**
`wildtype`	baseline	$\gamma \in [-3, 3]$	control
`hypodopamine`	DA=0.05, no phasic	—	MPTP, reserpine
`asd`	5-HT=0.85, OXT↓ social avoidance	sensory γ_{floor}=+1.2, integrative γ_{ceil}=−0.5	shank3, SERT mutant
`schizophrenia`	NMDA×0.3, DA=0.85	γ_{ceil}=−1.5 ($\pi \approx 0.18$)	DISC1, ketamine
`anxiety`	NA=0.75, flee thr=0.12	γ_{floor}=+1.0 ($\pi \geq 0.73$)	CRF overexpression
`depression`	DA=0.15, 5-HT=0.2	γ_{ceil}=−0.8 ($\pi \leq 0.31$)	CMS model
`adhd`	DA=0.35, ACh=0.25	—	DA deficit model
`ptsd`	amygdala×2, cortisol	γ_{floor}=+0.8 ($\pi \geq 0.69$)	trauma model

Precision-based disorder modeling. The two-compartment predictive coding architecture (Section 3) enables a principled account of precision abnormalities across disorders, following the "aberrant precision" framework of Adams et al. (2013). Each disorder sets structural bounds on the learnable precision parameter γ in all TwoCompColumn modules, producing persistent phenotypes that cannot be overwritten by online precision learning:

- **Schizophrenia**: $\gamma_{\text{ceiling}} = -1.5$ ($\pi \leq 0.18$) models NMDA hypofunction as a permanent structural deficit in precision. The brain attenuates prediction errors, producing a 3× reduction in total free energy and aberrant salience (Adams et al. 2013).

- **Anxiety**: $\gamma_{\text{floor}} = +1.0$ ($\pi \geq 0.73$) prevents precision from dropping below a hypervigilant baseline. The fish cannot habituate to non-threatening stimuli, over-weighting sensory prediction errors.

- **Depression**: $\gamma_{\text{ceiling}} = -0.8$ ($\pi \leq 0.31$) caps precision at a low level, modeling anhedonia as diminished responsiveness to prediction errors.

- **ASD**: Per-module uneven precision (Lawson et al. 2014). Sensory modules receive elevated γ_{floor} (detail-focused perception) while integrative modules (thalamus, pallium) receive reduced γ_{ceiling} (weak top-down integration).

- **PTSD**: $\gamma_{\text{floor}} = +0.8$ ($\pi \geq 0.69$), combined with elevated amygdala gain and cortisol, produces persistent hypervigilance with inability to down-regulate threat-related precision.

Testing disorder phenotypes. All disorder profiles can be run through the scenario suite by passing a modified personality and neuromodulatory initialization to any scenario function. Results are directly comparable across conditions because episode length, food distribution, and random seeds are held constant. Quantitative outcome measures—food intake, flee fraction, geo-coverage learning rate, $\mathbf{W}_{\text{LA}\to\text{CeA}}$ acquisition curve, social weight calibration rate—provide testable predictions at the group (population) and individual (per-episode) level.

5.5 Multi-Species Validation Platform (vzlab)

To demonstrate that the active inference framework generalises beyond zebrafish, we developed **vzlab**, a lightweight multi-species virtual laboratory that packages four invertebrate and vertebrate model organisms under a common four-tier validation protocol. Each species is implemented as a `BrainModule`–`Environment` pair registered in a central species registry; a single call to `run_full_validation(species)` executes all four tiers and returns a structured report.

Tier 1 Behavioural battery. Species-specific behavioural tests check that the avatar reproduces the defining locomotor or learning signature of the biological model organism (pass threshold > 0.5 on each metric).

Tier 2 Atlas correspondence. Population-level firing rates are compared to synthetic reference templates encoding the expected relative activation ordering across conditions, using Spearman rank correlation (pass threshold $r_s > 0.5$).

Tier 3 Lesion replication. Known biological interventions (region ablations) are applied and the resulting directional change in the target metric is verified against the published phenotype.

Tier 4 Robustness (graceful degradation). Biological neural circuits tolerate substantial cell loss without catastrophic failure (Rakic, 2002). We simulate neuron death by applying independent Bernoulli dropout to each sensory unit *before* the brain's step, so the brain genuinely makes decisions from degraded input. Behavioral output (motor speed $\times$ directional control) is measured at 10%, 30%, and 50% dropout; pass thresholds are $\geq$80%, $\geq$50%, and $\geq$20% retention respectively. A *robustness index* (RI) summarises the area under the retention curve, normalised to $[0, 1]$.

Table 10 summarises all four species. All achieve Grade A$^+$ (all four tiers passing).

Table 10: vzlab multi-species validation. All four species achieve Grade A$^+$ (T1–T4 all pass). T2 reports minimum Spearman r_s; T4 reports the robustness index (RI) and retention at 30% neuron dropout.

Species	**Model**	**T1**	**T2 (r_s)**	**T3**	**T4 RI**	**30% ret.**
C. elegans	302-n connectome	✓	1.0	✓	0.982	98.1%
D. rerio	100k-n SNN (zebrafish)	✓	1.0	✓	1.000	100%
D. melanogaster	Mushroom body (2145 n)	✓	1.0	✓	0.946	93.7%
X. laevis	Spinal CPG (150 n)	✓	1.0	✓	1.000	100%

Robustness profiles. The zebrafish and *Xenopus* CPG achieve perfect retention (RI = 1.0) because their locomotor circuits are self-sustaining: once initiated by sensory input, the CPG generates rhythm independently of ongoing sensory signals. *C. elegans* shows $\geq$ 96% retention even at 50% dropout because crawl speed is driven by internal motor neuron tone, while only the *direction* of chemotaxis is impaired consistent with the observation that sensory-deprived worms continue locomoting but lose chemotactic precision. *Drosophila* is most sensitive (RI = 0.946) because odour-guided approach requires sustained PN input to KC activity; this matches the known APL-dependent fragility of the mushroom body to olfactory deprivation. Critically, the zebrafish sensory cascade from photoreceptor and chemosensory signals through tectum, amygdala, and EFE goal selection to motor output retains full behavioral coherence at all tested dropout levels (RI = 1.0), supporting the claim that the cells-to-behavior hierarchy is inherently fault-tolerant.

Implementation notes. The *Drosophila* mushroom body implements the All-Pathway Loop (APL) feedback inhibitor enforcing $\leq$ 10% Kenyon cell sparsity (Lin et al., 2014). The *Xenopus* spinal CPG (Roberts and Bhatt, 1998; Li et al., 2006) models four neuron classes: Rohon-Beard (RB) sensory neurons, descending interneurons (dIN) as rhythm generators, commissural inhibitory interneurons (cIN) enforcing left-right anti-phase, and motor neurons (MN); dIN ablation stops swimming (-100% speed) and cIN ablation disrupts alternation by -45%. Full source code is available in the `vzlab/` package.

6 Discussion

6.1 Biological Plausibility

The virtual zebrafish demonstrates that the free energy principle provides a unified computational framework for the diverse functions of a vertebrate brain. Exteroceptive perception (predictive coding), interoceptive regulation (allostatic inference), goal selection (expected free energy), attention (precision optimization), learning (free energy gradient descent), and planning (generative model simulation) are all implemented as instances of the same variational objective applied at different levels of the neural hierarchy (Friston, 2010; Parr et al., 2022).

Several architectural features enhance biological plausibility. The two-compartment neuron model with distinct somatic and apical compartments follows proposals for cortical predictive coding circuits (Sacramento et al., 2018), where apical dendrites receive top-down predictions and somata integrate bottom-up evidence. The separation between reward-modulated feedforward learning (recognition model) and reward-independent feedback learning (generative model) reflects the fundamental distinction in predictive coding between perceptual inference and perceptual learning (Rao and Ballard, 1999; Clark, 2013). All learning rules—genomic pretraining, RPE-gated Hebbian plasticity, PE-driven anti-Hebbian feedback, and TD-based value learning—are local in both space and time, requiring no backpropagation through time.

The motor architecture follows the biological organization of zebrafish locomotion: spinal half-centre oscillators produce rhythmic swimming, Mauthner cells mediate fast escape reflexes, and cerebellar forward models predict sensory consequences of motor commands (Marques et al., 2018; Temizer et al., 2015). The amygdala fear circuit with persistent threat arousal reproduces the asymmetric rise/decay dynamics observed in zebrafish pallium during predator avoidance (Ahrens et al., 2013).

The 75% E / 25% I architecture is not arbitrary: the mammalian neocortex maintains a remarkably stable 4:1 excitatory-to-inhibitory ratio across layers and species (Carandini and Heeger, 2012). By enforcing this ratio in every EI layer, the model avoids the "signal death" pathology that arises in excitatory-only architectures, where cascaded layers amplify noise until deep layers fall silent. With inhibitory interneurons present, NMDA-mediated recurrent excitation is dynamically balanced by $GABA_A$-mediated feedback inhibition, producing realistic sparse activity patterns (∼10–15% active cells per layer) that are both metabolically efficient and information-theoretically optimal for population coding. The four-axis neuromodulatory system (DA, NA, 5-HT, ACh) goes beyond the common practice of using a single modulator to represent reward, reflecting the distinct anatomical origins and circuit targets of each system: mesolimbic DA modulates basal ganglia action selection and STDP consolidation, locus coeruleus NA mediates arousal and aversive memory consolidation (Sara, 2009), raphe 5-HT regulates patience and inter-trial intervals, and basal forebrain ACh gates attention and appetitive plasticity. This disaggregation produces qualitatively different behavioral profiles under fear (high NA, low DA) versus foraging (high ACh, moderate DA) versus exploration (high 5-HT, moderate everything) that are not achievable with scalar reward modulation alone.

6.2 Limitations

Several limitations constrain the current model. First, while the WTA goal-selector now receives reward-modulated Hebbian updates, it achieves sufficient confidence (> 0.4) to override analytic EFE only intermittently during early training; after sufficient episodes the analytic and spiking pathways are expected to converge. Second, the object classifier achieves 96.2% accuracy on gameplay data but may degrade in mixed scenes with overlapping entities, despite online ACh- and NA-gated fine-tuning from behavioral confirmation signals. Third, motor output is primarily driven by retinal bearing analysis (80–85%) with the reticulospinal voluntary pathway contributing 15%; as pallium-D STDP matures over training rounds, the RS contribution is expected to grow. Fourth, flee direction now uses the internal

Kalman-filtered predator model rather than ground-truth position, which may briefly degrade escape precision during first detection when the tracker has not yet converged. Fifth, the easy-foraging scenario (D) previously scored 42/100 due to a retinal food-bearing sign error: the right-eye centroid angular offset was applied with inverted polarity (`ang_R * 1.5` corrected to `-ang_R * 1.5`), causing the fish to orbit away from right-side food. The fix was applied and confirmed: D recovered to 76/100 at round 177 as continued STDP training consolidated the corrected motor pathway. The predator-charge scenario (B) improved from 30 to 100/100 following flee-offset re-tuning and predictive flee: the Kalman tracker's 8-step intercept prediction now triggers flee before the predator closes to contact range. Sixth, computational cost limits real-time simulation: 29 spiking modules at 50 substeps per behavioral step require approximately 4 minutes per 500-step episode on Apple M1 MPS; VAE training is batched every 10 steps to reduce this overhead by 10×.

6.3 Comparison with Existing Virtual Organisms

Compared to Spaun (Eliasmith et al., 2012), which models cortical cognitive tasks in isolation from sensorimotor control, the virtual zebrafish implements a complete perception-action loop in an ecological environment. Unlike the virtual rodent of Merel et al. (2019), which uses deep RL without biological neural architecture, our model employs spiking neurons with local learning rules grounded in active inference. The OpenWorm project (OpenWorm, 2014) targets cellular-level accuracy in *C. elegans* but has not achieved complete behavioral repertoires. The virtual zebrafish occupies a unique niche: it sacrifices single-neuron biophysical accuracy for whole-brain functional completeness, enabling the study of how multiple brain systems interact to produce adaptive behavior.

6.4 Biological Predictions

A key criterion for a computational model's scientific value is its ability to generate specific, falsifiable predictions about the biological system it represents. The virtual zebrafish makes the following predictions spanning single-neuron, circuit, behavioral, and social levels of analysis.

Prediction 1: DA depletion will impair consolidated motor–goal associations but not immediate reactive behavior. In our model, STDP consolidation is gated on DA > 0.55 or phasic bursts; low tonic DA therefore prevents long-run weight strengthening while leaving reactive EFE computation intact. The prediction is that dopamine-depleted larval zebrafish (e.g., with MPTP or reserpine treatment) should show preserved single-trial food approach and escape reflexes but degraded improvement across repeated exposures—consistent with the role of mesolimbic DA in procedural but not reactive learning (Schultz et al., 1997).

Prediction 2: Habenula lesion will increase net food intake under environmental uncertainty but decrease behavioral flexibility. Our ablation study found that removing habenular frustration signaling increased mean food intake by +22%. This is counterintuitive but follows from the frustration-driven goal-switching mechanism: in patchy environments, habenula-induced strategy switching carries an opportunity cost. The prediction is that habenula-lesioned zebrafish should outperform controls in uniform food distributions but underperform in environments that require switching from a depleted patch to a new location (Amo et al., 2014). This could be tested with a patch-depletion paradigm where food density decreases over time in one zone.

Prediction 3: Cholinergic blockade will impair food-category learning but not enemy-category learning. In our model, the classifier output layer is updated with ACh-gated Hebbian reinforcement for the food class and NA-gated reinforcement for the enemy class. Selective cholinergic antagonism (atropine) should therefore impair the progressive improvement in food detection across episodes without affecting the consolidation of predator identity representations. This dissociation is consistent with evidence for cholinergic specificity in appetitive versus aversive learning across vertebrates (Daw et al., 2011).

Prediction 4: Circadian phase will modulate risk tolerance in the EFE cost–benefit calculation. The circadian activity drive $\mathcal{A}$ in our model biases EFE toward foraging and exploration during peak activity phases (high $\mathcal{A}$) and reduces this bias during rest phases. The prediction is that larval zebrafish at peak circadian activity (ZT6–12, lights-on phase) should be willing to approach food at shorter predator-distance thresholds than fish at ZT18–24 (lights-off phase), independent of acute sensory evidence. This could be tested by presenting food near a stationary predator model at different Zeitgeber times and measuring approach latency.

Prediction 5: Bold fish will act as social noise absorbers, damping false-alarm cascades in heterogeneous shoals. In the virtual laboratory Social Learner scenario, bold fish (low w_{alarm}) diverge from shy fish (high w_{alarm}) within 5–8 episodes. Placing one bold fish in a shoal of four shy fish should measurably reduce the group false-alarm rate compared to a shoal of five shy fish, because the bold fish's low alarm weight propagates via Boids velocity coupling and reduces collective freeze duration. This predicts a heterogeneity advantage for mixed-personality shoals that can be tested by tracking group freeze response to a non-threatening visual stimulus (looming disk below predator threshold) in sorted shoals.

Prediction 6: NMDA hypofunction will shorten goal-persistence duration before spatial learning degrades. In the schizophrenia-like phenotype, thalamo-pallial NMDA reduction impairs working memory persistence. The model predicts that goal-persistence duration (number of consecutive steps with the same goal before switching) decreases in proportion to NMDA conductance reduction, independent of foraging success. This is testable with ketamine dose-response curves and goal-bout duration extracted from automated behavioral tracking.

Prediction 7: Social isolation during fear conditioning will impair precision calibration of the alarm weight. The precision-EMA update in SocialMemory requires a conspecific to validate alarm signals (predator-was-close within the 20-step horizon). Isolated fish have no social feedback, so w_{alarm} cannot converge. This predicts that socially isolated zebrafish show more volatile alarm responses (inconsistent freeze magnitude across trials) than shoal-reared fish—a prediction testable by comparing alarm response variance in individually-housed versus group-housed larvae exposed to identical predator cue sequences.

6.5 Comparison with Rule-Based and Deep RL Baselines

Beyond the architectural comparison with Spaun and the virtual rodent, it is informative to compare the virtual zebrafish against simpler control agents operating in the same environment. A purely reactive agent (flee when enemy pixels >0, forage otherwise, random turn for exploration) serves as a minimal behavioral baseline. In 10 evaluation episodes (500 steps, same seeds as the multi-seed result), the reactive agent survives 312 ± 89 steps and collects 1.8 ± 0.9 food items: substantially below the virtual zebrafish's 500 ± 0 steps and 11.1 ± 2.8 food items. This improvement is attributable to three architectural capabilities absent from the reactive agent: (1) olfactory gradient following that directs the fish toward food sources outside the visual field; (2) Kalman-filtered predator tracking that enables evasion of non-visible threats; and (3) predictive EFE that balances foraging urgency against starvation risk rather than responding only to current sensory evidence. The model demonstrates that the added biological realism of E/I balance, multi-axis neuromodulation, and STDP provides functional benefit in ecologically challenging environments.

6.6 Future Directions

Several directions for future work are evident. Most immediately, extended online RL training across 50+ rounds with 5 fish in the multi-agent curriculum will reveal whether the STDP weights converge to reliable sensorimotor associations and whether the WTA goal selector's Hebbian projection learns to predict goal-appropriate pallium-D patterns.

Virtual laboratory expansion. The most impactful near-term additions are: (1) standardized assay arenas implementing the Novel Tank Test, Light-Dark Preference, and Social Preference Test in exact protocol correspondence with published zebrafish behavioral literature, enabling direct quantitative comparison; (2) an empirical data comparison pipeline that ingests EthoVision or DeepLabCut CSV trajectories and computes behavioral similarity metrics (time-in-zone match, learning curve R^2, inter-individual variance ratio); (3) multi-seed statistical infrastructure that runs each scenario across $N \geq 10$ random seeds and reports mean ± SD alongside Cohen's d effect sizes between experimental and control conditions.

Disorder modeling extensions. The three disorder profiles presented—hypodopaminergic, ASD-like, and schizophrenia-like—represent first-pass parameter perturbations. More biologically grounded implementations would introduce (1) dynamic DA depletion simulating nigrostriatal degeneration (progressive STDP failure across training rounds); (2) oxytocin/vasopressin systems for social bonding and pair preference, currently absent but central to ASD biology; (3) cortisol (HPA axis) feedback causing hippocampal-analog shrinkage under chronic stress (reducing place-cell capacity over rounds); and (4) endocannabinoid-mediated fear extinction, enabling post-traumatic-stress-like and resilience-like phenotypes that differ in extinction rate.

Developmental trajectory. All current models represent adult fish. Larval zebrafish (5–7 dpf), the dominant experimental model due to brain transparency, differ substantially: fewer active neuromodulatory systems, simpler EFE with only two goal axes (approach/avoid), no social memory, and pre-myelinated motor pathways. Introducing a developmental staging parameter that gates module activation in the same order as biological maturation would allow the virtual laboratory to generate predictions about *when* specific cognitive capacities emerge—a question directly testable with calcium imaging across zebrafish development.

Cross-species scaling. The vzlab multi-species platform (Section 5.5) already demonstrates that the same active inference framework generalises to *C. elegans*, *Drosophila melanogaster*, and *Xenopus laevis* tadpole, all achieving Grade A validation under the same three-tier protocol. Future extensions include mouse (∼70M neurons, partial connectome) and human cortical organoids, testing whether the same computational architecture scales across orders-of-magnitude differences in brain size and whether disorder phenotypes exhibit homologous behavioral signatures across species. The current data (35 rounds, Table 4) show the expected non-monotonic early trajectory; FLEE time dropped from 60% (round 20) to 29% (round 35) following EFE re-tuning, and food intake remained above 2 items for post-fix rounds despite high variance.

Integration with real zebrafish whole-brain calcium imaging data (Ahrens et al., 2013; Portugues et al., 2014) could constrain the connectivity weights and validate the model's predicted functional anatomy. Specifically, the EFE-guided goal trajectories could be compared against activity trajectories in the dorsal pallium during prey-capture and predator-avoidance epochs. The model's prediction that DA-gated STDP selectively consolidates motor associations during phasic dopamine bursts is testable with simultaneous dopamine sensor imaging (dLight) and behavioral annotation.

Transfer to robotic zebrafish platforms (Marques et al., 2018) would test whether the active inference brain architecture produces naturalistic locomotion kinematics when driving a physical body with hydrodynamic tail actuation. Full multi-agent dynamics with five BrainAgent instances would replace the lightweight conspecific models, enabling study of distributed goal selection in a school and the emergence of collective predator-avoidance strategies from individually-learned representations. The vzlab multi-species platform (Section 5.5) has already extended the active inference architecture to *Drosophila* and *Xenopus*; future targets include mouse and human cortical organoids to test scaling across orders-of-magnitude differences in neuron count.

7 Conclusion

We have presented a spiking neural network model of the larval zebrafish brain comprising 48 modules and 7,316+ Izhikevich neurons with conductance-based synapses that implements the complete active inference loop. The architecture uses Izhikevich neurons with 7 cell types, 25% inhibitory interneurons in every layer for E/I balance, four-axis neuromodulation (DA, NA, 5-HT, ACh), and three-factor STDP with eligibility traces for biologically accurate temporal credit assignment. The model integrates predictive coding, active inference goal selection, spiking reinforcement learning, VAE world models, and Hebbian habit formation within a single sensorimotor system. Six complementary learning rules—all local and biologically plausible—operate simultaneously: (1) DA-gated three-factor STDP with 500 ms eligibility traces across the tectum–thalamus–pallium pathway; (2) PE-driven anti-Hebbian feedback weight learning; (3) TD-based RL critic with goal-persistence shortcutting on negative prediction error; (4) reward-modulated Hebbian learning of the WTA goal-selector projection; (5) ACh-gated food class reinforcement and NA-gated enemy class reinforcement in the classifier; and (6) online VAE ELBO minimization for latent world modeling. Key results (Table 12) include 84/100 decision rationality (round 177 checkpoint), perfect 10/10 seed survival (500-step episodes, 0 predator captures), olfaction and habenula identified as most impactful via ablation (−35% food each), and a 96.2% spiking classifier with online fine-tuning. The model demonstrates that the free energy principle provides a unified computational language sufficient to organize the diverse functions of a vertebrate brain—from retinal sampling through cerebellar forward models to circadian-gated behavioral regulation—using exclusively local, biologically plausible learning rules.

Table 11: The 48-module v2 architecture. SNN-Izh: Izhikevich spiking neurons. SNN-LIF: leaky integrate-and-fire. Rate: rate-coded continuous. Analytic: non-neural computation.

Module	Type	Cell Types	Neurons	Category
Retina (4 RGC types)	Rate	–	2 000	Sensory
Tectum (8 hemi. layers)	SNN-Izh	CH, IB, RS	3 200	Sensory
Classifier	SNN-LIF	LIF	128	Sensory
Lateral line	SNN-Izh	RS	20	Sensory
Olfaction	SNN-Izh	RS	20	Sensory
Color vision (4 cones)	SNN-Izh	RS	32	Sensory
Vestibular	SNN-Izh	RS	6	Sensory
Proprioception	SNN-Izh	RS	8	Sensory
Nociception (3 channels)	SNN-Izh	RS	6	Sensory
Auditory (Weberian)	SNN-Izh	RS	6	Sensory
Gustatory (3 taste ch.)	SNN-Izh	RS	6	Sensory
Tactile (somatotopic)	SNN-Izh	RS	6	Sensory
Hypothalamus (5 nuclei)	SNN-Izh	RS	10	Homeostatic
Pineal gland	SNN-Izh	RS	6	Homeostatic
Inferior olive	SNN-Izh	RS	8	Cerebellar
Vagus nerve	SNN-Izh	RS	10	Autonomic
Pituitary	SNN-Izh	RS	12	Endocrine
Area postrema	SNN-Izh	RS	8	Autonomic
NTS	SNN-Izh	RS	10	Autonomic
Lateral line efferent	SNN-Izh	RS	8	Sensory
Thalamus L+R (TC + TRN)	SNN-Izh	TC, LTS, FS	380	Relay
Pallium (S + D)	SNN-Izh	RS, IB	2 400	Cortex
Working memory	SNN-Izh	RS, FS	40	Cortex
Goal selector (WTA)	SNN-Izh	RS	4	Decision
Basal ganglia (D1/D2/GPi)	Rate	MSN	760	Action
Reticulospinal	Rate	–	42	Motor
Spinal CPG	SNN-LIF	V2a, V0d, MN	32	Motor
Amygdala (LA/CeA/ITC)	SNN-Izh	RS, IB, FS	50	Limbic

Table 12: v2 performance summary.

Metric	v2
SNN neurons	7 316+ (Izhikevich)
Cell types	7 (RS/IB/FS/CH/LTS/TC/MSN)
Inhibitory neurons	25% per layer
Neuromodulatory axes	4 (DA/NA/5-HT/ACh)
Modules	48 modules
Classifier accuracy	96.2%
Multi-seed survival (10)	500 ± 0 steps (10/10 full, 0 caught)
Decision rationality	84/100
Curriculum	3/3 PASS
Multi-agent	5 fish $\times$ full v2 brain
Personalities	bold/shy/explorer/social/default
Predator AI	5-state strategic hunting

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